Electricity Supply Shocks and Manufacturing: Evidence from the Fukushima Nuclear Accident

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We show that adverse shocks to electricity supply—an input accounting for only 2% of revenue—can cause substantial and persistent declines in industrial production. Using confidential establishment-level data and instrumenting with initial supply conditions, we find that shocks following the 2011 Fukushima nuclear accident reduced Japanese manufacturing electricity use by 6.4% and output by 1.8%, implying annual output losses of 5.3 trillion yen (US$44 billion). Establishments adjusted little through energy substitution, product switching, outsourcing, within-firm reallocation, or relocation. Our production function highlights two mechanisms that amplify output losses: establishments cannot easily substitute other inputs for electricity, and reduced electricity use leaves machinery and equipment temporarily idle. We quantify how electricity-market reforms, such as regional market integration, can mitigate these losses.

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1 Introduction

Modern economies run on electricity, yet its supply cannot be taken for granted, even in advanced economies. Russia's war in Ukraine and recurrent conflict in the Middle East illustrate what the International Energy Agency describes as "significant fragilities in today's global energy system" (IEA, 2024). Natural disasters pose additional risks to electricity supply. Electricity is now the world's largest single source of useful energy, and its role is set to grow (Figure 1). The stakes are rising: the more economies depend on electricity, the more they stand to lose when its supply is disrupted. Despite this growing dependence, little is known about how adverse shocks to electricity supply affect the industrial sector, a major consumer of electricity.[1]

This paper uses the sweeping changes in Japan's electricity policy following the 2011 Fukushima nuclear accident to study how sustained, nationwide electricity supply shocks affect manufacturing production. On March 11, 2011, a magnitude 9.0 earthquake and subsequent tsunami struck the Fukushima Daiichi nuclear power plant, triggering "the worst civil nuclear disaster since the one at Chernobyl in 1986" (UNSCEAR, 2021). The resulting reactor meltdowns and radioactive releases sparked widespread concerns about nuclear safety. The accident marked "a major turning point" in Japan's energy policy (ANRE, 2022), as the country responded by progressively taking nuclear reactors offline. Nuclear power's share of peak-day electricity supply collapsed from 17.4% in 2010 to 1.4% in 2013 (METI, 2016) and recovered to only 2.9% by 2018. This loss of nuclear supply triggered nationwide electricity shortages on an extraordinary scale—"perhaps one of the most severe electricity shortfalls in history" (IEA, 2011).

Our empirical strategy draws on confidential establishment-level data from Japan's Census of Manufactures for 2005–2018 and wide regional variation in electricity supply shocks. Although nuclear generation declined nationwide, the resulting supply shortfalls differed sharply across Japan's nine electricity service territories. The Kansai service territory, for example, had relatively little reserve capacity before the accident and immediately lost nuclear generation that had accounted for about 26% of its peak electricity supply. Despite efforts to expand generation from other sources, the region had to reduce electricity demand by as much as 23% at times to avoid blackouts.[2] The government sought to achieve these demand reductions largely through moral suasion. Replacing nuclear power with other energy

[1] A few studies examine how changes in electricity taxes or tariffs affect firm performance in advanced economies (Gerster and Lamp, 2024; von Graevenitz and Rottner, 2024). Others study blackouts or greyouts in developing economies (Fisher-Vanden et al., 2015; Allcott et al., 2016; Cole et al., 2018; Gonzales et al., 2026). A broader literature studies oil and other energy shocks (Kilian, 2008; Hamilton, 2013).

[2] Electricity supply must meet demand in real time, including at its peak. Nuclear power provided approximately 26% of Kansai's peak-day supply in 2010.

sources, particularly expensive thermal generation, also drove up electricity prices. In contrast, regions with nuclear shares below 10% or large reserve margins, such as Chugoku and Hokkaido, faced relatively mild electricity supply shocks. As such, we exploit this regional variation using instruments based on pre-accident nuclear reliance and reserve margins.

The electricity supply shocks caused substantial and persistent contractions in manufacturing activity. After 2013, when Japan's nuclear shutdown looked set to last, the shocks reduced electricity use by 6.4%, labor use by 1.1% and materials use by 1.6%, and output by 1.8%. These contractions persisted through 2018, seven years after the accident. The 1.8% reduction in output implies an annual loss of approximately 5.3 trillion yen (US$44 billion),[3] equivalent to roughly three years of manufacturing growth at the sector's pre-accident average annual real growth rate. Neither direct damage from the earthquake and nuclear accident nor supply-chain disruptions nor reductions in local demand can explain these effects on manufacturing.

The shocks induce little adjustment along margins that could mitigate output losses. Electricity supply shocks do not increase establishments' use of other energy inputs in production. The shocks also have little effect on outsourcing or product switching, two ways establishments could reduce the electricity intensity of production. Nor do they meaningfully shift production toward establishments in less affected regions or induce establishments to relocate. Instead, they mainly reduce operating hours and scale back production. Nonetheless, our interpretation is that establishments act to limit output losses when feasible. The roughly 4% of establishments that own generators do, in fact, increase on-site electricity generation in response to the shocks. For most establishments, however, options for mitigating output losses appear limited or costly.

These empirical findings raise two related questions: why can shocks to electricity—an input with a small expenditure share—generate such large output losses, and how can policies mitigate these losses? In our data, electricity expenditure accounts for only 2% of establishment revenue, yet a 6.4% reduction in electricity use is associated with a 1.8% reduction in output, far exceeding a standard factor-share benchmark.[4] We have demonstrated one contributing factor—establishments have limited ability to cushion the shocks—but this alone does not fully explain the magnitude of these output losses.

To explain such large output losses, we estimate a production function and use it to predict input and output responses to electricity supply shocks. We incorporate two mechanisms that amplify output losses: limited substitutability between electricity and other

[3]Throughout the paper, we convert yen to US dollars at 120 yen per dollar.

[4]We estimate electricity's output elasticity at 0.015. Holding other inputs and productivity fixed, a 6.4% decline in electricity use therefore implies an output loss of only $0.015 \times 6.4\% = 0.096\%$, far below our estimated decline of 1.8%.

inputs and electricity-induced capital idleness. Limited substitutability makes it difficult to replace electricity with other inputs. Capital idleness operates differently: the shocks reduce the utilization of existing capital and the services it provides, even after accounting for substitution among inputs (Finn, 2000). For example, a steel establishment may be unable to fully utilize its electricity-dependent furnaces when electricity availability declines or when shocks reduce operating hours. The resulting capital idleness directly reduces output and indirectly amplifies the contraction by lowering the productivity of other inputs.

We estimate the model by matching its predicted responses of electricity use, labor, materials, and output to electricity supply shocks with the corresponding reduced-form estimates. Under a CES production technology, we estimate an elasticity of substitution of 0.57 between electricity and other inputs. If we instead assume greater substitutability, the model predicts substantially more substitution away from electricity. The model predicts that the shocks lead to a larger reduction in electricity use but smaller reductions in materials use and labor input, and therefore in output—a pattern inconsistent with the reduced-form estimates. We also estimate a capital-idleness elasticity of 0.17. In an average region, the shocks reduce electricity use by 6.4%. This leaves about 1.1% (6.4% $\times$ 0.17) of installed capital idle, reducing capital services by 1.1%. This decline in capital service may appear modest, yet it substantially amplifies output losses because capital plays a much larger role in production than electricity does directly. This capital idleness mechanism increases the predicted output loss from 0.6% to 1.7%, nearly tripling the losses.

How can policies mitigate the output losses caused by electricity supply shocks? We found that manufacturing establishments adapted little to the shocks, so we focus on policies that reduce the severity of the shocks they face. Japan used moral suasion to curb electricity demand, while replacing lost nuclear power with fossil-fuel generation raised electricity prices. Without moral suasion, our production function model still predicts that higher electricity prices reduce output by about 1.5%—roughly 90% of the combined impact. These results suggest that removing barriers to lower electricity prices in Japan could substantially mitigate output losses.

We quantify how electricity-market reforms, particularly regional market integration, can mitigate these losses by lowering electricity prices. Japan has long recognized that its fragmented electricity market creates inefficiencies (ANRE, 2017). Indeed, in testimony before Japan's House of Representatives, a government official argued that regional market integration could "make it possible to offer consumers lower electricity rates" (House of Representatives, 2014). Without integration, high-cost generators may operate in one region to meet local demand while lower-cost generators elsewhere sit idle, keeping electricity prices unnecessarily high. These inefficiencies were especially costly in our setting, as each region had to rely more heavily on high-cost generators within its own territory to replace lost

nuclear supply and meet local demand. We estimate that regional market integration lowers the average log electricity price by 2.8 log points and reduces price-induced output losses by 32.3% relative to the baseline without integration.

This paper makes several contributions. First, we provide the first micro-level evidence on how adverse shocks to electricity supply affect industrial production in an advanced economy. Decades of research have examined the economic effects of oil and natural gas shocks (Hamilton, 1983; Rotemberg and Woodford, 1996; Kilian, 2008; Hamilton, 2013). Yet oil and natural gas account for less than one-quarter of global electricity generation (IEA, 2026), raising questions about whether these findings generalize to electricity supply shocks. A substantial body of research has also examined electricity supply disruptions, focusing largely on outages in developing economies. (Fisher-Vanden et al., 2015; Allcott et al., 2016; Cole et al., 2018; Fried and Lagakos, 2023).[5] A smaller literature examines how electricity prices affect firm performance in advanced economies (Kahn and Mansur, 2013; Bijnens et al., 2022), including studies exploiting changes in electricity taxes, tariffs, and incentives (Gerster and Lamp, 2024; von Graevenitz and Rottner, 2024).[6] We instead examine large, persistent electricity supply shocks—an increasingly important concern as economies electrify—and document substantial output losses.

Second, we contribute to a longstanding literature on why shocks to inputs with small expenditure shares can cause large output losses. Standard economic theory predicts that small shocks to inputs with small cost shares should have limited effects on output (Solow, 1957; Hulten, 1978). Empirical evidence suggests otherwise: oil and intermediate-input shocks can generate output losses far exceeding these predictions (Rotemberg and Woodford, 1996; Barrot and Sauvagnat, 2016; Boehm et al., 2019). Previous studies propose several amplification mechanisms, including limited input substitution and reduced capital utilization (Atkeson and Kehoe, 1999; Finn, 2000; Atalay, 2017; Baqaee and Farhi, 2019). Yet much of the existing work relies on aggregate data that obscure how individual producers adjust or on calibrated models that do not directly estimate key production parameters. We instead use establishment-level data and find that manufacturers adapted little to the shocks. Exploiting quasi-experimental variation in electricity supply, we further show that limited input

[5]Several studies of blackouts document substantially larger effects than those we estimate. For example, Allcott et al. (2016) estimate that electricity shortages reduce establishment revenues in India by 5–10%, roughly three to six times our estimated output decline of 1.8%. For Sub-Saharan Africa, Cole et al. (2018) estimate that reducing outages to South African levels would increase firm sales by 85.1%. Although the shocks and outcomes differ across settings, the smaller effects in our setting may partly reflect a different margin of adjustment: establishments continue to operate at higher marginal costs, whereas blackouts can halt production.

[6]A substantial literature also examines how households respond to electricity prices and conservation policies, including during electricity crises (Reiss and White, 2008; Leighty and Meier, 2011; Ito, 2015; Costa and Gerard, 2021; He and Tanaka, 2023).

substitution and capital idleness amplified output losses far beyond the standard factor-share benchmark.

This paper also contributes to research on nuclear phase-outs and their economic consequences.[7] Major nuclear accidents have repeatedly reshaped energy policy worldwide. After Chernobyl, Italy exited nuclear power; after Fukushima, Germany accelerated its nuclear phase-out, Japan rapidly shut down nuclear generation, and countries such as Switzerland and South Korea adopted or considered policies to reduce their reliance on nuclear power. A growing literature shows that nuclear shutdowns can impose sizable costs by removing low-marginal-cost generation and increasing reliance on fossil fuels. These costs include higher electricity prices (Davis and Hausman, 2016), health damages from increased air pollution (Severnini, 2017; Jarvis et al., 2022), and increased mortality from reduced household electricity consumption (Neidell et al., 2021; He and Tanaka, 2023). We extend this literature by studying a less-examined downstream effect: industrial production. We show that the abrupt shift in nuclear policy harmed industrial production by raising electricity prices.

2 Institutional Background

2.1 Japan's Regulated Electricity Markets

During our study period, Japan's electricity sector was heavily regulated, generating wide regional variation in electricity supply shocks following major disruptions to the electricity system. Japan's mainland has nine regional electricity markets. A single traditional utility dominated each market, supplying more than 90% of local electricity, while wholesale trading was negligible (JEPIC, 2019). In particular, three features of Japan's electricity system contributed to regional variation in these supply shocks.

First, regions differed substantially in their electricity supply mix and reserve margins—two factors that contributed to regional variation in supply shocks. In 2010, nuclear power accounted for 17.4% of total peak-day electricity supply across the nine mainland regions (ESDV, 2012). Nuclear power played a relatively large role partly because Japan imports nearly all the fossil fuels it consumes and the government had long promoted nuclear generation to strengthen energy security. Nuclear reliance nevertheless varied substantially across regions because reactor siting depended on access to cooling water, compliance with seismic safety requirements, and local political acceptance. The nuclear share ranged from 0% in Chugoku and 9.2% in Chubu to 25.6% in Kansai and 29.1% in Shikoku, as shown in Panel A

[7] Existing studies document the short- and long-term impacts of nuclear accidents and related disasters. Such events can disrupt supply chains (Carvalho et al., 2021), affect health (Almond et al., 2009; Danzer and Danzer, 2016), alter perceptions of nuclear risk and induce anxiety (Fu et al., 2026), and reduce housing and land prices (Kawaguchi and Yukutake, 2017).

of Figure 2. Electricity reserve margins also varied across regions. In 2010, Hokkaido and Shikoku had margins of 16.4% and 17.7%, providing larger buffers against the loss of nuclear supply. By contrast, Kansai and Tokyo had margins of only 5.7% and 6.9%, leaving them more exposed to electricity shortages following the nuclear shutdown.

Second, regulation limited interregional transmission and trade, preventing local electricity supply shortfalls from being smoothed across regions. Under the Electricity Business Act, each utility could monopolize electricity supply within its service area but had a legal obligation to maintain reliable local supply (Government of Japan, 1964). Before the accident, most regions traded little electricity with one another (Panel A of Figure 2). During severe electricity supply shocks, exporting electricity became even more difficult because doing so could jeopardize supply to utilities' own customers. Following the Fukushima nuclear accident, interregional electricity transfers indeed remained extremely small, often below 1% of peak-day demand (see Section 4.1).[8]

Third, under Japan's regulatory system, changes in electricity supply costs translated directly into changes in regional electricity prices. Japan used cost-of-service regulation, under which utilities charged customers prices that covered electricity supply costs plus a regulated return. Replacing nuclear power with more expensive thermal generation therefore raised retail prices, especially in regions with greater pre-accident nuclear reliance. Retail electricity prices rose with a delay because of lengthy regulatory approval processes.

2.2 The Fukushima Nuclear Accident and Nuclear Phase-Out

The Fukushima nuclear accident was, in the words of UNSCEAR (2021), "the worst civil nuclear disaster since the one at Chernobyl in 1986." The disaster began with the March 2011 Great East Japan Earthquake—Japan's largest recorded earthquake and the world's fourth-largest since 1900. The earthquake and ensuing tsunami killed more than 15,000 people (U.S. DOI, 2011; USGS, 2011). Tsunami waves 14–15 meters high hit the Fukushima Daiichi nuclear power plant and disabled its cooling systems (IAEA, 2015). As a result, several reactor cores melted down, releasing large amounts of radioactive material. This accident immediately changed public and regulatory attitudes toward nuclear safety.

The accident led to sweeping changes in Japan's electricity policy. It immediately changed public and regulatory attitudes toward nuclear safety and intensified opposition to nuclear power. Japan gradually shut down its nuclear reactors, and all were offline by May 2012. The government subsequently imposed safety standards so stringent that most reactors could

[8] Although physical constraints existed, most notably the incompatibility between eastern Japan's 50 Hz grid and western Japan's 60 Hz grid, the primary barrier to equalizing supply shocks was institutional rather than technological.

not meet them and remained offline for years. Nuclear power thus nearly disappeared from Japan's electricity supply—a shift later described as "a major turning point in Japan's energy policy" (ANRE, 2022). Nuclear power's share of peak supply was just 1.4% in 2013 and remained extremely low at 2.9% in 2018 (METI, 2016; OCCTO, 2018).

The nuclear shutdown triggered a nationwide electricity supply crisis by removing a source that had supplied 17.4% of peak-day electricity in 2010. In mid-2011, the IEA described Japan's situation as "perhaps one of the most severe electricity shortfalls in history" (IEA, 2011).

To address the electricity supply shocks, policymakers adopted both demand- and supply-side measures. On the demand side, the government used moral suasion to maintain electricity-system reliability and prevent widespread blackouts.[9] The government encouraged manufacturers to improve energy efficiency, dim lighting, reduce air-conditioning use, adjust work schedules, and shift operations away from peak hours (Kakuchi, 2011). On the supply side, the government eased regulatory requirements to facilitate additional thermal generation. Utilities reactivated idle fossil-fuel plants, increased output from existing thermal generators, and added new thermal capacity. Replacing nuclear generation with higher-marginal-cost thermal generation raised electricity generation costs, which were largely passed through to consumers as higher retail prices. Regions that relied more heavily on thermal generation to replace lost nuclear output consequently faced larger price increases.

Panels B and C of Figure 2 illustrate the consequences of the Fukushima nuclear accident for Japan's electricity markets. Panel B shows the sharp and persistent decline in nuclear supply. Increased supply from thermal plants only partially offset this loss, leaving total peak-day electricity supply below its pre-accident level. Panel C shows that electricity prices rose through 2014 and remained elevated through 2018, despite a temporary decline. Manufacturers therefore faced persistently higher electricity input costs, with substantial variation across regions.

3 Data

3.1 Peak-Day Electricity Demand and Supply

We assemble regional-level data on peak-day electricity demand and supply from 2005 to 2018 to measure electricity supply shocks. Our primary data source is the Electricity Supply–Demand Verification Subcommittee (ESDV, 2012),[10] a government body that convenes an-

[9] In summer 2011, the government mandated a 15% reduction in peak electricity use relative to 2010, but only for large establishments in the Tokyo and Tohoku service areas (METI, 2011).

[10] We supplement these data with information from the Agency for Natural Resources and Energy (ANRE) (ANRE, 2022) and utilities' power supply planning documents.

nually to assess summer and winter electricity supply conditions. The subcommittee collects detailed information on regional supply capacity and demand on peak days, which the national government uses to design and implement interventions in electricity markets. The government used 2010 as the baseline for its demand-reduction policies and assessed supply against peak-day rather than peak-hour demand.

3.2 Census of Manufactures

We use establishment-level administrative data from the Census of Manufactures (CMF) for 2005–2018, compiled by the Ministry of Economy, Trade, and Industry (METI) (METI, 2012). The census annually surveys all manufacturing establishments with 30 or more employees, covering approximately 80% of total manufacturing output in Japan. Appendix A-1.1 provides further details on the survey coverage and sample selection.

The CMF provides detailed establishment-level information on outputs, inputs, and production characteristics. We measure output using total revenue and inputs using expenditures, with all values adjusted for taxes and deflated. Because the CMF does not report establishment-level electricity prices, we deflate electricity expenditures using average regional prices. This approach is likely reasonable because regional utilities largely set prices through regulated tariff schedules (JEPIC, 2019). We use the three-digit classification as our baseline definition of an "industry" unless otherwise noted. Appendix A-1.2 provides further details on variable construction.

We construct the analysis sample using three main restrictions. First, we exclude establishments located in municipalities severely affected by the Great East Japan Earthquake and the Fukushima nuclear accident, using an affected-area definition consistent with Carvalho et al. (2021). Second, for each outcome regression, we restrict the sample to establishments observed in both the baseline year, 2010, and the outcome year, so that the dependent variable, the change in log outcomes relative to 2010, is defined. Third, we restrict the CMF sample to establishments with nonmissing and positive values for output and primary input expenditures, including labor, materials, electricity, and capital. We winsorize these output and input variables at the 1st and 99th percentiles to mitigate the influence of outliers.

The CMF includes several additional features useful for our analysis. Firm identifiers allow us to link establishments owned by the same firm and study whether activity shifts across regions within multi-establishment firms. A subset of establishments reports product-level information, enabling us to analyze product switching. We also measure establishment entry and exit using aggregate data from METI. Appendix A-1.3 describes the construction of these measures.

3.3 Energy Consumption Statistics

We complement the CMF with establishment-level administrative data from the Energy Consumption Statistics (ECS) provided by ANRE ((Agency for Natural Resources and Energy, 2199)). These surveys report physical quantities of fuel inputs, including natural gas (city gas, LPG, and LNG), gasoline, kerosene, heavy oil, diesel oil, and steam. We convert all fuels into a common unit based on their heat content to construct a standardized measure of energy use. A key advantage of the ECS is that it distinguishes fuel used directly in production from fuel used for on-site electricity generation through boilers, cogeneration systems, or diesel generators. Note that the ECS and CMF use different sampling frames. Appendix A-2 provides additional details.

3.4 Additional Datasets

We supplement the establishment-level data with additional regional data for robustness checks. To account for supply-chain disruptions from the earthquake and tsunami, we use prefecture-by-industry input-output tables from RIETI (RIETI, 2005). We also collect prefecture-level data on local economic conditions, including gross prefectural product, population, construction investment, consumption, and labor- market conditions (MIC, 2025). We also collect prefecture-level data on manufacturing working hours per worker, defined as the sum of scheduled and overtime hours, from the Ministry of Health, Labour and Welfare's Monthly Labour Survey (MHLW, 2005–2018). See details in Appendix A-3.

3.5 Summary Statistics

Appendix Table A1 presents summary statistics. In the CMF sample (N = 448,148), the average establishment generates approximately 5,122 million yen in annual output (about US$43 million at 120 yen per dollar) and employs 128 workers. Approximately 48% of establishments belong to multi-establishment firms. On average, labor, capital, and materials account for 22%, 3.3%, and 51% of revenue, respectively, while electricity represents a relatively small share at 2.0%. In the ECS sample (N = 373,218), generator ownership averages 5.8% among observations with known status. On average, establishments purchase 16.4 TJ of electricity from the grid and generate 2.2 TJ on site.

4 Empirical Strategy

4.1 Constructing a Measure of Electricity Supply Shocks

We construct a measure of regional electricity supply shocks ($\mathcal{S}$) based on peak-day supply-demand conditions. Peak days are central to blackout risk because electricity supply must

meet demand in real time, including on the highest-demand days.

We define shock intensity as the gap between required and available supply in a given region and year, normalized by pre-accident peak demand. Intuitively, it measures the demand reduction needed to avoid large-scale blackouts in that region during that year.

Specifically, we define the shock as:

$$\mathcal{S}_{rt} = \frac{1}{D^{\text{peak}}_{r,2010}} \Big(\underbrace{1.05\, D^{\text{peak}}_{r,2010}}_{\text{required supply}} - \underbrace{S^{\text{plan,peak}}_{rt}}_{\text{available supply}} \Big) \cdot \mathbb{1}\left\{ S^{\text{plan,peak}}_{rt} < 1.05\, D^{\text{peak}}_{r,2010} \right\}. \tag{1}$$

where $D^{\text{peak}}_{r,2010}$ denotes realized demand on the 2010 peak day in region r. Required supply equals this pre-accident peak-day demand plus a 5% reserve margin. Because weather and economic activity can cause unexpected fluctuations in electricity demand, the government recommends that regional utilities maintain a 5% reserve margin (ESDV, 2012). $S^{\text{plan,peak}}_{rt}$ is planned available supply capacity for the peak day in year t. Available supply declines when existing nuclear capacity is phased out and increases when utilities take supply-side measures, such as restarting retired thermal plants or building new ones. The indicator sets the shock to zero in region-years in which available supply exceeds required supply.

Japanese regions faced severe electricity supply shocks, but their intensity varied widely across regions. On average during 2011–2018, manufacturing establishments faced electricity supply shocks of about 7.7%. Nonetheless, Panel A of Figure 3 shows wide geographic variation, with Tokyo and Kansai facing the most severe shocks. Panel B shows a sharp rise in average shock intensity after the 2011 disaster. Shocks persisted through 2018, with substantial variation across regions and over time.

Figure B1 shows each region's planned peak-day supply by energy source and required supply in 2013 and 2015. In 2013, only Kansai included nuclear power in its supply plan, and all regions fell short of required supply. Interregional transmission only partly offset these shortfalls. In 2015, shocks persisted in eight of the nine regions, and none included nuclear power in their supply plans.

4.2 Conceptual Framework

We use a conceptual framework that guides our empirical analysis. Manufacturing establishments produce output (Y) using capital (K), labor (L), materials (M), and electricity (E), together with Hicks-neutral productivity Ω. The production function is

$$Y = \Omega F(K, L, M, E), \tag{2}$$

where $F(\cdot)$ is differentiable. For this benchmark, suppose establishments take output and input prices as given. Differentiating $\ln Y$ with respect to electricity supply shock severity

$\mathcal{S}$ yields

$$\frac{d\ln Y}{d\mathcal{S}} = \theta_E \frac{d\ln E}{d\mathcal{S}} + \theta_M \frac{d\ln M}{d\mathcal{S}} + \theta_L \frac{d\ln L}{d\mathcal{S}} + \theta_K \frac{d\ln K}{d\mathcal{S}} + \frac{d\ln \Omega}{d\mathcal{S}}, \tag{3}$$

where $\theta_X \equiv \partial \ln F(\cdot)/\partial \ln X$ denotes the output elasticity with respect to input $X \in \{K, L, M, E\}$. The effect on log output therefore equals the sum of the effects on log inputs, weighted by their output elasticities, plus the effect on log productivity. Appendix B-2 provides the detailed derivation.

This framework provides a benchmark for the output impacts of electricity supply shocks. Consider a small, transitory, and unexpected shock. Holding labor, materials, capital, and productivity fixed at L^*, M^*, K^*, and Ω^*, respectively, the effects are approximated by

$$\left.\frac{d\ln Y}{d\mathcal{S}}\right|_{L=L^*,\, M=M^*,\, K=K^*,\, \Omega=\Omega^*} \approx \theta_E^* \frac{d\ln E}{d\mathcal{S}}. \tag{4}$$

Electricity expenditure accounts for only about 2% of revenue on average, implying a small electricity output elasticity, θ_E^*. The benchmark therefore predicts modest output losses from electricity reductions alone.

This benchmark, however, may not apply to the shocks we study. Electricity supply shocks after the Fukushima accident were large and predictable, and they persisted for years. Under such circumstances, establishments could re-optimize materials, labor, and, to some extent, capital.

4.3 Econometric Specification

We estimate the effects of electricity supply shocks on the outcomes in equation (3), including electricity use, labor, materials, output, and productivity. Using output as an example, we estimate the following specification:

$$\Delta \ln Y_{ijrt} = \rho_1 \mathcal{S}_{rt} \mathbb{1}\{2011 \leq t \leq 2012\} + \rho_2 \mathcal{S}_{rt} \mathbb{1}\{2013 \leq t \leq 2018\} + \mu_{jt} + \epsilon_{ijrt}, \tag{5}$$

where $\Delta \ln Y_{ijrt} = \ln Y_{ijrt} - \ln Y_{ijr,2010}$ is the change in establishment-level log output relative to 2010, the year before the nuclear accident, for establishment i in industry j, region r, and year t. This baseline is consistent with our shock measure, defined relative to 2010 electricity use (ESDV, 2012). The variable $\mathcal{S}_{rt}$ denotes the intensity of the local electricity supply shock, μ_{jt} denotes industry-by-year fixed effects, and ϵ_{ijrt} is the error term.

We separately estimate the impacts of electricity supply shocks in two periods: ρ_1 for the phase-out period (2011–2012), when Japan began shutting down nuclear reactors and policy uncertainty remained high, and ρ_2 for the post-phase-out period (2013–2018), when Japan tightened reactor restart requirements, making restarts substantially more difficult and less likely.

We also estimate the following event-study specification to examine pre-trends and dynamic effects:

$$\Delta \ln Y_{ijr\tau} = \sum_{t \neq 2010} \beta_t \mathbf{1}\{\tau = t\} \mathcal{S}_{r\tau} + \mu_{j\tau} + \varepsilon_{ijr\tau}, \tag{6}$$

where τ indexes calendar years in the sample. Statistically insignificant estimates of β_t in pre-accident years support the assumption that high- and low-shock regions followed similar outcome trajectories before the accident. The post-accident coefficients describe how the relationship between realized supply shocks and the outcome evolves over time. We run the regression separately for each year $t \neq 2010$ to estimate β_t. This specification allows us to obtain both OLS and 2SLS event-study estimates (see Section 4.4). Because electricity supply shocks vary over time, we use each region's mean shock for the pre-accident period.

We conduct inference using the wild cluster bootstrap, with regions as clusters. Because treatment assignment varies across only nine regions, conventional cluster-robust asymptotic approximations may be unreliable and prone to over-rejection (Cameron and Miller, 2015). Following Canay et al. (2021) and Roodman et al. (2019), we use Rademacher weights to calculate wild-bootstrap Wald p-values and confidence intervals. Tables report standard errors clustered by region, with significance stars based on the bootstrap p-values.

4.4 Instrumenting for Electricity Supply Shocks

Regional electricity supply shocks may be endogenous because of post-accident policy responses. Some regions reactivated dormant thermal plants or restarted nuclear reactors. These supply-side adjustments may be correlated with unobserved local economic conditions, such as recovering industrial demand or political influence.

To address this concern, we use instruments based on regions' pre-accident nuclear-generation shares and reserve margins. Regions with greater nuclear dependence faced larger potential supply losses, while those with lower reserve margins had less capacity to absorb them. Our instrument vector, $\mathbf{Z}_{rt}$, is:

$$\mathbf{Z}_{rt} = \left(Sh^{\text{nuclear}}_{r,2010} \times g^{\text{nuclear}}_{t,-r},\ Sh^{RM}_{r,2010} \right). \tag{7}$$

Here, $Sh^{\text{nuclear}}_{r,2010}$ denotes the 2010 nuclear-generation share in region r, and $g^{\text{nuclear}}_{t,-r}$ is the national growth rate of nuclear generation. The second instrument, $Sh^{RM}_{r,2010}$, denotes region r's reserve margin in 2010. All variables are normalized by 2010 peak-day demand to match the scale of the shock measure.

4.5 Production Function Estimation

4.5.1 CES Production Function

We estimate a constant-elasticity-of-substitution (CES) production function, assuming that establishments are price takers in output and input markets. For establishment i in year t, output is given by

$$Y_{it} = \exp(\omega_{it} + \epsilon_{it}) \left(\alpha_K K_{it}^{\rho} + \alpha_L L_{it}^{\rho} + \alpha_M M_{it}^{\rho} + \alpha_E E_{it}^{\rho}\right)^{1/\rho}, \tag{8}$$

where ω_{it} denotes establishment-level log productivity and ϵ_{it} is an idiosyncratic ex post production shock. The elasticity of substitution is $\sigma \equiv 1/(1-\rho)$, and the share parameters are positive and normalized such that $\alpha_K + \alpha_L + \alpha_M + \alpha_E = 1$.

Taking logs, equation (8) becomes

$$y_{it} = f(k_{it}, l_{it}, m_{it}, e_{it}) + \omega_{it} + \epsilon_{it}, \tag{9}$$

where lowercase variables denote logs and $f(\cdot)$ is the log of the CES aggregator.

We adopt the following timing assumptions. Establishments first choose the quasi-fixed inputs, labor and capital, before observing current-period productivity, ω_{it}. Productivity follows a first-order Markov process,

$$\omega_{it} = h(\omega_{it-1}) + \eta_{it}, \tag{10}$$

where η_{it} satisfies $\mathbb{E}[\eta_{it} \mid k_{it}, l_{it}] = 0$. After observing ω_{it}, establishments choose the flexible inputs, electricity and materials, conditional on their predetermined labor and capital. Finally, after all input choices have been made, establishments experience the ex post shock ϵ_{it}, which was unanticipated at the time of those choices and satisfies $\mathbb{E}[\epsilon_{it} \mid k_{it}, l_{it}, m_{it}, e_{it}, \omega_{it}] = 0$.

4.5.2 Estimation Strategy

We recover the CES parameters in two steps. First, we estimate industry-specific output elasticities for each input (see details in Appendix B-3.1). We identify the output elasticities of the flexible inputs, electricity and materials, using static first-order conditions following Gandhi et al. (2020), and estimate those of the quasi-fixed inputs, labor and capital, using a control-function approach (e.g., Olley and Pakes, 1996; Ackerberg et al., 2015). Second, conditional on a candidate elasticity of substitution (σ), we recover the CES distribution parameters (α) from the estimated output elasticities (see details in Appendix B-3.2). Under cost minimization, an input's normalized output elasticity equals its share of total input cost. The CES first-order conditions then map these cost shares into the distribution parameters.

This approach offers three advantages. First, it allows us to estimate a gross-output production function using static first-order conditions. As Gandhi et al. (2020) show, stan-

dard proxy-variable approaches generally cannot identify flexible-input output elasticities in gross-output production functions without additional variation in flexible-input demand, which is often unavailable. Second, the first step delivers output elasticities, so the implied CES share parameters can be recovered for alternative values of σ without re-estimating the output elasticities. Third, estimating the model separately by industry allows production technology to differ across industries.

4.5.3 Production Function Estimation and Results

We estimate the structural parameters of the CES production function separately for 152 three-digit manufacturing industries using pre-accident data from 2005–2010.

Columns (1)–(3) of Table 1 summarize the distribution of industry-specific output elasticities estimated using the method of Gandhi et al. (2020). Across industries, the estimated elasticities average 0.439 for materials, 0.015 for electricity, 0.358 for labor, and 0.144 for capital. Electricity output elasticities vary substantially across industries, with the 75th percentile (0.018) more than twice the 25th percentile (0.008).

OLS yields larger elasticities for flexible inputs, consistent with previous findings (Gandhi et al., 2020). Column (4) of Table 1 reports an OLS electricity elasticity of 0.066, more than four times our preferred estimate of 0.015, and a materials elasticity of 0.485, compared with 0.439. These differences are consistent with upward simultaneity bias. Establishments increase flexible inputs in response to positive productivity that OLS does not observe, leading OLS to attribute part of the resulting output increase to those inputs. For quasi-fixed inputs, the pattern is mixed: OLS understates the capital elasticity (0.042 versus 0.144) but overstates the labor elasticity (0.431 versus 0.358).

5 Electricity Supply Shocks and Establishment Outcomes

5.1 Trends in Inputs and Outputs of Manufacturing Establishments

We present trends in manufacturing inputs, output, and productivity among establishments in high- and low-shock regions. We classify regions as high-shock if their mean post-accident shock intensity is at or above the median. Figure 4 plots these outcomes over 2005–2018 as log changes relative to their 2010 levels.

Figure 4 shows that, following the 2011 accident, manufacturing inputs and output declined in high-shock regions relative to low-shock regions. In Panel A, electricity use followed similar trends in the two groups before 2011. After the accident, however, electricity use fell substantially in high-shock regions but changed only modestly in low-shock regions. This divergence widened as nuclear reactors were gradually shut down during 2011–2012

and persisted throughout 2013–2018. Employment, wage bill, material inputs, and revenue (Panels B–D and F) show similar, though smaller, divergences. By contrast, productivity (Panel E) shows no comparable divergence.

5.2 First-stage Results: Testing the Instruments

We assess instrument relevance by estimating the first-stage relationship between regional electricity supply shocks and our two instruments: predicted changes in nuclear generation and the initial reserve margin. We regress local shock intensity on both instruments.

Predicted losses in nuclear generation intensify local electricity supply shocks. Table 2 reports estimated coefficients on predicted nuclear supply growth of -0.58 for 2011–2012 (Column (1)) and -0.26 for 2013–2018 (Column (2)). In the average region, nuclear generation accounted for 18% of 2010 peak-day demand, and these reactors went almost completely offline during 2013–2018. The estimated coefficient implies that the nuclear phase-out increased shock intensity by about 4.5 percentage points on average—the additional reduction in peak demand needed to avoid blackouts.

The initial reserve margin mitigates local electricity supply shocks. Table 2 reports estimated coefficients of -1.39 for 2011–2012 (Column (1)) and -0.83 for 2013–2018 (Column (2)), consistent with our conjecture that pre-existing capacity buffers helped regions absorb these shocks.

The first stage is strong despite having only nine independent clusters. Predicted nuclear supply growth and the initial reserve margin are jointly statistically significant at the 5% level in both periods. The first-stage F-statistic is 173 for 2011–2012 and 28 for 2013–2018.[11]

5.3 The Effects of Supply Shocks on Manufacturing Outcomes

5.3.1 Impacts on Electricity Use

We estimate how electricity supply shocks affect manufacturing inputs, productivity, and output. Figure 5 presents event-study estimates from Equation (6). For each outcome, we run separate annual regressions of its log change relative to 2010 on local electricity supply shocks, using OLS and 2SLS. The 2SLS regressions use predicted nuclear-generation losses and initial reserve margins as instruments.

Panel A of Figure 5 shows that electricity supply shocks led to a persistent decline in manufacturing electricity use. Before the accident, the estimated effects are statistically indistinguishable from zero. This pattern suggests similar pre-trends in regions with high

[11] These values exceed the conventional threshold of $F > 10$ suggested by Stock and Yogo (2003). Lee et al. (2022) argue that substantially larger F-statistics may be desirable in some settings to ensure robust instrument relevance.

and low supply shocks. Following the accident, the shocks reduced manufacturing electricity use, and the effect grew over time. The effect was small in 2011 and 2012, when Japan's nuclear policy was still in transition. It increased in 2013 and persisted through 2018. The OLS and 2SLS estimates are similar in magnitude and statistical precision.

Our estimates suggest that electricity supply shocks reduced manufacturing electricity use by about 6.4% on average during 2013–2018. We calculate this effect using the 2013–2018 2SLS coefficient of −0.87 (Table 3, Panel B, Column (1)) and the mean shock intensity of 7.4%. In a high-shock region such as Tokyo, where shock intensity averaged 10.9%, the implied reduction in manufacturing electricity use was about 9.5%. In contrast, immediately after the accident, the estimated effects were small and statistically insignificant.

5.3.2 Impacts on Labor and Materials

Figure 5 shows that electricity supply shocks reduce employment (Panel B), wage bill (Panel C), and materials (Panel D), and that these effects grow over time. Across these outcomes, we do not find systematic differences in pre-trends and the OLS and IV estimates are similar in magnitude.

On average, electricity supply shocks reduced employment and wage bill by about 1.1% each and material inputs by about 1.6% during 2013–2018. Columns (2)–(4) of Table 3 report the estimated effects on these outcomes. The IV estimates in Panel B imply that a 1 percentage point increase in the supply shock reduced employment and wage bill by about 0.15% each and material inputs by about 0.22%. Despite having only nine regional clusters, we find statistically significant effects on employment and wage bill at the 10% level.

Our analysis by employment type shows that electricity supply shocks disproportionately reduce employment among non-regular workers, such as part-time and short-term contract workers (see Appendix A-1.2 for definitions). Appendix Table C1 shows little effect on wages per worker. The estimated coefficient for the post-phase-out period (2013–2018) is 0.00 (Panel B, Column (1)), suggesting that establishments adjusted employment rather than wages. The estimated employment effects are similar for male and female workers: −0.13 in Column (2) and −0.14 in Column (3), respectively. However, the effects differ substantially between regular and non-regular workers. The coefficient for regular workers is small and statistically insignificant at −0.01 (Column (4)), whereas the coefficient for non-regular workers is −0.93 (Column (5)). These results indicate that electricity supply shocks disproportionately harm non-regular workers who often face greater job insecurity.

5.3.3 Impacts on Productivity

Electricity supply shocks have small and statistically insignificant effects on TFP. Panel E of Figure 5 shows that the estimated effects of electricity supply shocks on TFP remain small and statistically insignificant over time. The 2SLS point estimates are −0.05 for 2011–2012 and −0.07 for 2013–2018 (Table 3, Panel B, Column (5)). In theory, electricity supply shocks can either raise or lower productivity. On the one hand, shocks may encourage manufacturers to raise productivity by streamlining production processes or adopting more efficient production technologies (Martin et al., 2012). On the other hand, historical experience suggests that shocks to input markets may reduce productivity (Kilian, 2008). Regardless of these competing predictions, we find only small and statistically insignificant effects on productivity.

In Appendix C-2, we find little evidence that electricity supply shocks induced electricity-saving technological change. This finding may seem surprising given the substantial efforts Japanese manufacturers made to conserve electricity after the accident (Kimura and Nishio, 2016). Following the Fukushima nuclear accident, electricity supply shocks reduced electricity use more than labor inputs. This decline may reflect higher electricity prices relative to wages or improvements in electricity productivity relative to labor productivity. Our results suggest that higher electricity prices relative to wages explain much of this decline during 2013–2018.

5.3.4 Output Impacts and Aggregate Losses

Electricity supply shocks reduced manufacturing output, with effects that persisted through 2018—seven years after the accident (Panel F of Figure 5). While the estimated effects during 2011–2012 were small, electricity supply shocks reduced manufacturing output by about 1.8% on average during 2013–2018. We calculate this effect by multiplying the 2SLS estimate of approximately −0.248 in Table 3 (Panel B, Column (6)) by the mean shock intensity of 7.4%.

The estimated output losses far exceed what the benchmark predicts. Under the benchmark in Section 4.2, a 6.4% reduction in electricity use would reduce output by only about 0.1%, holding other inputs and productivity fixed, because electricity's estimated output elasticity is only 0.015 ($0.015 \times 6.4\% = 0.096\%$). Our estimated output loss of 1.8% is nearly 19 times this benchmark.

Back-of-the-envelope calculations suggest substantial aggregate economic losses from electricity supply shocks. Applying the estimated 1.8% decline during 2013–2018 to Japan's 2010 manufacturing output of 289 trillion yen (METI, 2012) yields an annual loss of approximately 5.3 trillion yen, equivalent to roughly US$44 billion. For comparison, this loss corresponds

to nearly three years of pre-crisis manufacturing growth: according to the CMF, real manufacturing output grew by an average of approximately 0.6% per year between 1995 and 2007 (METI, 2012). Furthermore, applying the estimated 1.14% reduction in employment to the sector's 2010 workforce of 7.7 million implies approximately 88,000 fewer workers employed each year during 2013–2018.[12]

Three caveats qualify our aggregate estimates. First, our counterfactual assumes that each region has enough planned peak-day electricity supply to meet its 2010 peak demand with a 5% reserve margin. Second, our regional estimates may not fully capture broader economic effects. Electricity supply shocks may also affect less-exposed regions through supply chains, input and output prices, and exchange-rate adjustments. Our analysis does not account for these effects, which may amplify or offset the estimated losses. Third, we extrapolate our estimates to the entire manufacturing sector, although our analysis includes only establishments with at least 30 employees.

We benchmark our estimates against previous studies of electricity and broader energy-market shocks. Our estimated decline in manufacturing output is broadly comparable in magnitude to the effects of historical oil shocks. A large literature links major energy-market disruptions to contractions in economic activity, while emphasizing that the magnitude depends on the source of the shock and the empirical specification (Kilian, 2008; Hamilton, 2013). Kilian and Vigfusson (2017) estimate that oil-price shocks associated with the Iranian Revolution and Iraq's invasion of Kuwait reduced U.S. real GDP by 0.5–3.5% over the following two years, depending on the model specification. Recent estimates also suggest that disruptions to Russian energy supplies reduced German GDP by 0.5–3% in the short run (Bachmann et al., 2024).

In contrast, studies exploiting changes in electricity taxes, tariffs, and incentives in advanced economies generally find small or statistically insignificant effects on output and competitiveness (Wolverton et al., 2022; von Graevenitz and Rottner, 2024; Gerster and Lamp, 2024). As discussed in Section 4.2, one explanation is that establishments do not reoptimize other inputs in response to modest, temporary electricity price changes, limiting the resulting effects on output. These findings may not generalize to large adverse shocks to electricity markets, as we find significant output losses in this setting.

Evidence from developing countries shows larger output losses from grayouts and blackouts (Fisher-Vanden et al., 2015; Allcott et al., 2016; Cole et al., 2018; Fried and Lagakos, 2023). For example, Allcott et al. (2016) estimate that these disruptions reduce Indian manufacturers' annual revenues by 5–10%, while Cole et al. (2018) estimate that reducing power

[12]The employment total covers manufacturing establishments with at least four employees (METI, 2012). We assume that the post-phase-out estimate applies to aggregate manufacturing employment in 2010 and use the unrounded coefficient to calculate the employment effect.

outages to South African levels would increase firm sales by 85.1%. These studies differ from ours in how they measure electricity disruptions and firm outcomes, so the estimates are not directly comparable. Nevertheless, they suggest far larger output losses than those we find in Japan, plausibly because grayouts constrain production and blackouts can halt it entirely. Japanese manufacturers generally remained in operation but faced persistently higher electricity prices and government pressure to reduce electricity demand through moral suasion.

5.4 Heterogeneous Impacts

The heterogeneity analysis sheds light on the channels through which electricity supply shocks affect manufacturing outcomes. We examine whether the effects vary with establishment- and industry-level electricity and capital intensity, measured in 2010.[13] For each measure, we split the sample at the median, estimate Equation (5) separately for the two groups, and report the 2SLS estimates for the complete-shutdown period.

The effect of electricity supply shocks on electricity use varies little with baseline electricity or capital intensity, measured at either the establishment or industry level. Panel A of Figure 6 shows that the subgroup estimates generally lie close to the full-sample estimate of -0.87. For each sample split, we fail to reject equality between the estimates for the high- and low-intensity groups. This limited heterogeneity is consistent with the broad reach of Japan's policy response, which did not target establishments with specific characteristics. Establishments within the same region generally faced common moral suasion and electricity price increases.

By contrast, the output impacts vary sharply with capital intensity. Panel B of Figure 6 reports a coefficient of -0.24 for capital-intensive establishments, compared with -0.05 for establishments with low capital intensity. The corresponding coefficient is -0.37 for establishments in capital-intensive industries, compared with 0.03 for those in less capital-intensive industries. We find no comparable heterogeneity by electricity intensity. These patterns are consistent with electricity shortages reducing capital utilization and amplifying output losses at capital-intensive establishments (Berndt and Wood, 1975; Finn, 2000). We incorporate this mechanism into the production technology in Section 6.3.1.

[13] Electricity intensity is electricity expenditure divided by revenue, and capital intensity is the capital stock divided by revenue.

5.5 Alternative Explanations and Robustness Checks

5.5.1 Supply-Chain Disruptions and Local Demand Shocks

We examine whether supply-chain disruptions provide an alternative explanation for our finding that electricity supply shocks led to large output contractions. Previous research documents that damage from the Great East Japan Earthquake propagated through input–output linkages (Carvalho et al., 2021). Such disruptions could therefore have contributed to the estimated output declines.

We find that controlling for exposure to supply-chain disruptions does not materially alter our main estimates. We account for direct and total exposure to supply-chain disruptions caused by the earthquake and nuclear accident. Direct exposure measures the share of intermediate inputs sourced from disaster-affected prefectures, while total exposure also captures dependence on these areas through suppliers' suppliers (see definitions in Appendix A-3). In Table C3, adding these controls separately or jointly yields 2SLS coefficients for 2013–2018 ranging from -0.89 to -0.88 for electricity use and from -0.23 to -0.22 for output (Columns (2)–(4)). These estimates are close to the baseline coefficients of -0.87 and -0.25, respectively (Column (1)).

5.5.2 Local Demand Control

Table C4 examines whether our baseline estimates change when controlling for changes in local demand conditions relative to 2010, including economic activity outside manufacturing, population, consumption, residential construction investment, nonmanufacturing employment, and consumer prices. The 2SLS estimates for 2013–2018 remain broadly stable as these controls are added cumulatively. For example, controlling for gross prefectural product in the primary and tertiary sectors changes the electricity-use coefficient from -0.87 to -0.85, while the output coefficient remains -0.25 (Columns (1) and (2)). With all controls included, the coefficients for electricity use and output are -0.88 and -0.25, respectively (Column (6)), close to the baseline estimates of -0.87 and -0.25. These results suggest that differences in observed local demand conditions do not explain our main findings.

5.5.3 Sensitivity to Individual Regions

We re-estimate the baseline regressions excluding one electricity region at a time. Because mainland Japan has only nine electricity regions, a single region could disproportionately influence the baseline estimates.

Our results show that no single region alone drives the baseline results. Table C5 shows that the estimated effects during 2013–2018 remain negative and broadly similar in magnitude across exclusions. The 2SLS coefficients range from -1.01 to -0.60 for electricity

use and from −0.32 to −0.17 for output. For instance, excluding Chubu (Region 5 in Figure 3) reduces the magnitude of the electricity-use coefficient from −0.87 to −0.70 and the output coefficient from −0.25 to −0.17. On the other hand, excluding Hokkaido (Region 1) increases the magnitude of the electricity-use coefficient to −1.01, while excluding Kansai (Region 6) increases the magnitude of the output coefficient to −0.32. Yet the estimates remain negative and broadly comparable in magnitude to the baseline estimates across all exclusions.

5.6 Establishment Responses to Electricity Supply Shocks

This section examines how and to what extent manufacturing establishments adjust their behavior in ways that could mitigate output losses from electricity supply shocks.

Overall, electricity supply shocks induced little adjustment along margins that could mitigate output losses. We consider product choice, outsourcing, production reallocation across establishments, relocation, and entry and exit, but find little evidence of adjustment along these margins. These findings support our model in Section 4.2, which abstracts from these adjustment margins. We report the detailed results in Appendix D.

First, we examine product switching and outsourcing, two potential ways for establishments to reduce the electricity intensity of their production processes. However, we find little evidence that electricity supply shocks affected either margin (Table 4). We also find no convincing evidence that establishments reallocated production across regions or relocated in response to the shocks (Tables D1 and D2). Furthermore, we find no evidence that establishments increased their use of other energy inputs (Table 5, Column (1)). Most of the estimated effects are not only statistically insignificant but also small in magnitude.

One notable exception is on-site generation, which helped cushion output losses but was available to few establishments. The shocks substantially increased on-site generation among establishments with pre-existing generators (Table 5, Column (3)). These establishments showed no statistically significant decline in output (Figure D1), suggesting that on-site generation helped mitigate the shocks. However, only about 4% of establishments in our sample owned generators, limiting the scope of this adaptation strategy.

Instead, the shocks appear to have reduced manufacturing working hours per worker. We estimate a 0.8% decline at the mean shock intensity during 2013–2018 (Table 4, Panel B, Column (4)), which could account for approximately 42% of the estimated 1.8% output loss if output declines proportionally with working hours.

Why did manufacturing establishments make so few adjustments to mitigate output losses from electricity supply shocks and instead reduce working hours and output? Several commonly proposed explanations appear unlikely in our setting. First, insufficient time to adapt is unlikely to explain the limited responses. Our data cover seven years after the Fukushima

accident, giving establishments considerable time to adjust their production processes and invest in alternative technologies. Second, firms likely did not forgo adaptation simply because they expected the shocks to be temporary. By September 2012, the Japanese government had announced a goal of phasing out nuclear power over the following two decades, casting doubt on a return to pre-accident nuclear generation (EEC, 2012). Third, firms' widespread awareness of electricity supply risks suggests that information frictions played a limited role. In a 2013 survey, 70% of manufacturing firms anticipated production losses under continued electricity supply uncertainty and conservation requirements (Keidanren, 2013). Fourth, financial constraints are unlikely to explain establishments' limited investment. Although some firms faced credit constraints immediately after the 2011 earthquake, lending conditions generally improved and remained favorable during 2013–2018, as indicated by the Bank of Japan's Tankan survey (Bank of Japan, 2018).[14]

Instead, our findings suggest that establishments recognized the need to adapt and had incentives to do so, but lacked affordable ways to respond. The experience of establishments with pre-existing generators illustrates this point. These establishments, accounting for about 4% of our sample, increased on-site electricity generation and avoided substantial output losses. Yet installing new generation capacity requires substantial upfront investments and operating costs that most establishments may struggle to justify. We find little evidence of other effective adaptation strategies, which may partly reflect the difficulty establishments faced in finding viable alternatives. More broadly, investments in adaptation may have been particularly difficult to justify in Japan. Manufacturing employment in Japan fell by roughly one-third, from 14.6 million in 1990 to 9.6 million in 2010, while firms increasingly moved production overseas to access cheaper labor and growing foreign markets (Kawaguchi, 2013; Kang and Piao, 2015). Against this backdrop, firms may have had weaker incentives to invest in adapting their existing domestic plants. Overall, our findings highlight the challenges manufacturing establishments face in adapting to large and persistent electricity supply shocks.

6 From Reduced-Form Effects to Production Mechanisms

This section uses the production function estimated in Section 4.5 to interpret the reduced-form effects of electricity supply shocks. We predict the impacts of electricity supply shocks using this production function and compare these predictions with the reduced-form esti-

[14]The Tankan diffusion index (DI) for financial institutions' lending attitudes, defined as the percentage of firms reporting accommodative lending attitudes minus the percentage reporting severe attitudes, increased from 8 to 21 for small enterprises and from 20 to 25 for large enterprises between December 2013 and December 2018. These figures cover all industries (Bank of Japan, 2013, 2018).

mates. This comparison sheds light on the mechanisms that could reproduce the reduced-form responses

6.1 Simulation Framework

We use the production function to predict electricity use, materials, labor, and output with and without the observed electricity supply shocks. The resulting differences correspond to the reduced-form effects estimated using Equation (3). We use the same sample as in the reduced-form analysis. We use 2010 as the baseline year and average conditions over 2013–2018 to represent the post-phase-out period.

Each plant i is characterized by its 2010 productivity, $\omega_{i,2010}$; output, $Y_{i,2010}$; and generator-ownership status, $G_{i,2010}$. For each industry j, we use the estimated output elasticities to construct the baseline cost shares s_{hj} for $h \in K, L, M, E$. For each candidate elasticity of substitution, σ, we hold these cost shares fixed and recover the corresponding CES distribution parameters, α_{hj}, as described in Appendix B-3.2.

We assume that each plant faces isoelastic demand,

$$Y_i = B_i P_i^{-\eta}, \tag{11}$$

where B_i is a plant-specific demand shifter and $\eta > 1$ is the absolute value of the price elasticity of demand. We calibrate B_i so that demand at the baseline price reproduces the plant's observed output in 2010. We set $\eta = 5$, which lies within the range of sector-specific elasticities of substitution estimated for U.S. manufacturing by Shapiro and Walker (2018).

Electricity supply shocks affected manufacturing plants through two channels: quantity rationing and increases in the electricity price, p^E. To model the rationing channel, we follow He and Tanaka (2023) and use the government's regional electricity-saving targets, denoted by $\overline{R}_r$. We impose each region's target on its manufacturing plants during the 90-day shock season, consistent with the government's demand-reduction strategy (ESDV, 2012). The plant-weighted target averages 7.7% in our sample. To model the price channel, we use only the price changes attributable to local supply shocks because observed electricity prices also reflect global energy prices, which contributed to the price increase in 2014 and the declines in 2015–2016. Appendix E-2 shows that a 1-percentage-point increase in shock intensity raises the log electricity price by approximately 1.12%. At the mean shock intensity, this estimate implies that the average plant faced an electricity price increase of approximately 8.4% during the post-phase-out period.

For each plant, we use its production technology and demand to predict profit-maximizing input use and output with and without the supply shocks. We hold technology and demand fixed across the two scenarios and vary only electricity prices and the rationing constraint. Because rationing applies only during the peak season, we solve the plant's optimization

problem separately for the peak and off-peak seasons and aggregate the predictions to obtain annual outcomes. Table E1 summarizes the simulation inputs, and Appendix E-3 describes the procedure.

The simulation also allows responses to differ by access to self-generation. In Section 5.6, we show that plants with generators reduce their purchases from grid utilities but exhibit no meaningful declines in other inputs or output. Based on this evidence, we assume that these plants offset reductions in grid purchases through self-generation and maintain their baseline levels of non-electricity inputs and output.

6.2 Cobb-Douglas Technology

We first predict the impacts of electricity supply shocks assuming $\sigma = 1$, that is, a standard Cobb–Douglas benchmark. We use the model to predict the effects on electricity use, labor, materials, and output and compare these predictions with the reduced-form effects evaluated at the mean shock intensity in Section 5.3.

The Cobb–Douglas benchmark fails to reproduce the reduced-form impacts in two respects. First, it substantially understates the effect of electricity supply shocks on output. As shown in Figure 7, the model predicts an output decline of only 0.6% (Panel B), compared with the reduced-form estimate of 1.7% (Panel A). Second, it misrepresents the relative responses of production inputs. The model predicts a 9.0% decline in electricity use, 2.4 percentage points larger than the reduced-form estimate of 6.6%, while predicting declines of only 0.4% in labor and materials, compared with reduced-form estimates of 1.3%. By imposing a unit elasticity of substitution, $\sigma = 1$, the Cobb–Douglas specification allows plants to substitute away from electricity too easily. It therefore overstates the decline in electricity use while understating the contractions in other inputs and output.

6.3 Amplification Mechanisms

6.3.1 Capital Idleness

We augment our model to address two discrepancies between the reduced-form estimates and the Cobb–Douglas predictions. First, to explain why the Cobb–Douglas model understates output losses, we allow electricity supply shocks to idle installed capital. Consider a steel plant that operates electric furnaces. When electricity is unavailable, the plant must reduce furnace operating hours, leaving part of its capacity idle. This capital idleness directly reduces output and lowers the marginal products of labor and materials, prompting the plant to reduce both inputs. Capital idleness thus amplifies the output loss beyond the direct effect of reduced electricity use.

We introduce capital idleness for two reasons. First, the literature on energy shocks

establishes capital utilization as an important mechanism underlying large fluctuations in aggregate output (see, e.g., Berndt and Wood, 1975; Finn, 2000). Electricity unavailability can similarly amplify output losses by reducing capital utilization. Second, the heterogeneity results in Section 5.4 support this mechanism. Electricity supply shocks cause disproportionately large output losses at more capital-intensive plants and in more capital-intensive industries, whereas their effects do not vary systematically with electricity intensity. This pattern is consistent with capital utilization as the underlying mechanism.

We therefore modify the model so that production depends on effective capital services, $\widetilde{K}$, rather than directly on the installed capital stock, K:

$$Y_{it} = \Omega_i \left[\alpha_K \widetilde{K}_{it}^{\rho} + \alpha_L L_{it}^{\rho} + \alpha_M M_{it}^{\rho} + \alpha_E E_{it}^{\rho}\right]^{1/\rho}. \tag{12}$$

In this augmented model, electricity affects output directly as a production input and indirectly through capital utilization. For example, an electricity constraint can reduce the services provided by a steel plant's electric furnaces without changing its installed furnace capacity.

We formalize this mechanism by defining effective capital services as

$$\widetilde{K}_{it} = \left[(1-\kappa)K_{it}^{\rho} + \kappa\left(v_i E_{it}\right)^{\rho}\right]^{1/\rho}, \qquad v_i \equiv \frac{K_{i,2010}}{E_{i,2010}}, \tag{13}$$

where K_{it} is installed capital, $v_i E_{it}$ is the amount of capital that plant i can operate with its electricity input, and $\kappa \in [0,1]$ governs the dependence of effective capital services on electricity. The normalization $v_i \equiv K_{i,2010}/E_{i,2010}$ implies full capital utilization in 2010 and yields

$$\left.\frac{\partial \ln \widetilde{K}_{it}}{\partial \ln E_{it}}\right|_{t=2010} = \kappa. \tag{14}$$

Using effective capital services ($\widetilde{K}_{it}$) instead of installed capital (K_{it}) changes the baseline input cost shares. The Appendix E-4 describes how we incorporate capital idleness into the simulation. After this adjustment, the baseline capital and electricity cost shares are

$$\widetilde{s}_{Kj}(\kappa) = (1-\kappa)s_{Kj}, \qquad \widetilde{s}_{Ej}(\kappa) = s_{Ej} + \kappa s_{Kj}. \tag{15}$$

where s_{Kj} and s_{Ej} are the baseline capital and electricity cost shares in industry j, and $\widetilde{s}_{Kj}(\kappa)$ and $\widetilde{s}_{Ej}(\kappa)$ are the corresponding shares after accounting for electricity-dependent capital utilization.

A positive κ therefore shifts κs_{Kj} from the capital cost share to the electricity cost share. Electricity affects output both as a direct production input and by determining how much installed capital can operate. Consequently, higher electricity prices or tighter rationing have stronger effects on unit costs, input demand, and output when capital idleness is more important.

The magnitude of κ is ambiguous a priori. Some machinery may require electricity in nearly fixed proportions, leading electricity shortages to idle substantial capital. However, retrofitting machinery or changing operating practices may limit capital idleness. We therefore estimate κ from the data.

6.3.2 Input Complementarity

Second, to explain why the Cobb–Douglas model overstates the decline in electricity use and understates the declines in labor and materials, we relax its assumption that the elasticity of substitution, σ, equals one. When $\sigma < 1$, electricity is complementary to other inputs, so electricity supply shocks induce plants to reduce labor and materials rather than substitute these inputs for electricity. This limited substitutability produces a smaller decline in electricity use and larger declines in labor, materials, and output, bringing the model predictions closer to the reduced-form estimates.

6.3.3 Calibration

We jointly estimate the elasticity of substitution, σ, and the utilization elasticity, κ, using minimum-distance GMM. The empirical targets are the post-phase-out 2SLS impacts of electricity supply shocks on electricity use, labor, materials, and output. For each candidate pair (σ, κ), we calculate the model-implied responses using the simulation procedure described above. We then construct four moment discrepancies as the differences between the model-implied and reduced-form responses. We select the parameter pair that minimizes the sum of the squared discrepancies. Appendix E-4 presents the detailed procedure.

We obtain $\widehat{\sigma} = 0.57$ and $\widehat{\kappa} = 0.17$, which minimize the discrepancy between the model-implied and reduced-form impacts of electricity supply shocks. At the mean post-phase-out supply shock, the reduced-form estimates imply declines of 6.6% in electricity use, 1.3% in labor, 1.3% in materials, and 1.7% in output. The calibrated model predicts corresponding declines of 6.7%, 1.4%, 1.4%, and 1.6%. The model therefore closely reproduces the input and output responses.

The estimate $\widehat{\sigma} = 0.57$ indicates limited substitutability between electricity and other inputs. It implies stronger input complementarity than a Cobb–Douglas technology and falls within the range of approximately 0.2–0.8 reported for substitution between energy and other inputs.[15] Comparisons with estimates specific to electricity are more difficult because

[15] Using U.S. manufacturing-plant data and variation in energy prices, Hawkins-Pierot and Wagner (2026) estimates a common elasticity of substitution among capital, labor, and energy of 0.26 across all industries and 0.24 among energy-intensive industries. van der Werf (2008) reports energy substitution elasticities of approximately 0.16–0.62 across industries, while Bretschger and Jo (2024) estimates labor–energy elasticities of 0.61–0.80 using data on French manufacturing firms.

the empirical evidence remains scarce.[16]

Panel D of Figure 7 varies σ while holding $\kappa = 0.17$ and the baseline cost shares fixed. Lowering σ from its calibrated value of 0.57 to 0.2 strengthens input complementarity. The predicted decline in electricity use falls from 6.7% to 4.4%, while the declines in labor and materials increase from 1.4% to 2.0% and the output loss increases from 1.6% to 2.0%. Conversely, raising σ to 1.5 increases input substitutability. The predicted decline in electricity use rises to 13.8%, while the declines in labor and materials fall to 1.0% and the output loss falls to 1.4%.

The estimate $\widehat{\kappa} = 0.17$ implies that, at the baseline, a 1% reduction in electricity use reduces effective capital services by 0.17%. This utilization channel is quantitatively important because electricity's direct output elasticity is only 0.015, approximately one-tenth of capital's output elasticity of 0.139. Panel E of Figure 7 illustrates the extent to which capital idleness amplifies the effects of electricity supply shocks. Eliminating the utilization channel by setting $\kappa = 0$ reduces the predicted output loss from 1.6% to 0.6%, a reduction of approximately 61%. The corresponding declines in labor and materials fall from 1.4% to 0.5%. Conversely, setting $\kappa = 0.5$ increases the predicted output loss from 1.6% to 3.3%, more than doubling the effect. The corresponding declines in labor and materials increase to 2.9%.

7 Policy Counterfactuals

In this section, we use the structural model to evaluate counterfactual policies that could mitigate the output effects of electricity supply shocks.

7.1 Decomposing the Effects of Pricing and Rationing

Electricity supply shocks affected manufacturing plants through higher electricity prices and peak-season rationing. We use the structural model to decompose the effects of these two channels during the post-phase-out period (2013–2018). Their relative importance indicates whether policies should focus on reducing electricity costs or relaxing physical supply constraints.

To separate the price and rationing channels, we compare plant-level predictions for electricity use, labor, materials, and output across three scenarios: the 2010 no-shock baseline,

[16] Aggarwal (2026) estimates labor–electricity elasticities of approximately 0.1 in India's formal manufacturing sector and 0.2–0.5 in its informal manufacturing sector.

a price-only scenario, and a combined price-and-rationing scenario. The decomposition is

$$\underbrace{\ln Z_i^{\text{sim}}(\overline{p}_r^E, \overline{R}_r) - \ln Z_i^{\text{sim}}(1,0)}_{\text{Total impact}} = \underbrace{\ln Z_i^{\text{sim}}(\overline{p}_r^E, \overline{R}_r) - \ln Z_i^{\text{sim}}(\overline{p}_r^E, 0)}_{\text{Rationing contribution}} + \underbrace{\ln Z_i^{\text{sim}}(\overline{p}_r^E, 0) - \ln Z_i^{\text{sim}}(1,0)}_{\text{Price contribution}}. \tag{16}$$

where $Z_i^{\text{sim}}(\cdot,\cdot)$ denotes plant i's simulated annual outcome, with the first argument representing the relative electricity price and the second representing peak-season rationing intensity. We normalize the 2010 baseline to $(1,0)$. For region r, $\overline{p}_r^E$ is the predicted post-phase-out relative price, and $\overline{R}_r$ is the mean electricity-saving target used as the rationing intensity.

The first term on the right-hand side measures the rationing contribution by comparing the combined and price-only scenarios. The second term measures the price contribution by comparing the price-only scenario with the no-shock baseline. By construction, the two contributions sum to the total impact. This ordering assigns their interaction to the rationing contribution.

Table 6 shows that pricing accounts for most of the model-implied decline. During the post-phase-out period, the structural model predicts a 6.72% decline in electricity use. The price contribution is 6.23 percentage points, or 92.7% of the total decline, while the rationing contribution is 0.49 percentage points, or 7.3%. The model also predicts declines of 1.39% in labor, 1.39% in materials, and 1.57% in output. Pricing accounts for 1.30, 1.30, and 1.46 percentage points of these declines, respectively, corresponding to 93.0% of each total effect. Rationing accounts for the remaining 7.0%. Although the physical-rationing constraint binds for 85.7% of plants during summer, it applies for only 90 days per year and is generally small in magnitude. Consequently, rationing accounts for only about 7% of the annualized effects on electricity use, labor, materials, and output.

These results indicate that the nuclear shutdown primarily affected manufacturing plants through higher electricity prices. Utilities replaced nuclear generation with more expensive fossil generation, raising generation costs. Binding quantity constraints played a much smaller role. This pattern differs from the residential sector, where non-price interventions can meaningfully reduce electricity demand during scarcity (e.g., Reiss and White, 2008; He and Tanaka, 2023).

7.1.1 Elasticity Benchmarks

We report elasticities with respect to electricity-price changes using both the reduced-form estimates and the structural model. These elasticities provide benchmarks for the counterfactual analysis.

The reduced-form analysis implies an own-price elasticity of -0.80 for electricity, cross-price elasticities of -0.15 for both labor and materials, and an output elasticity of -0.20 (Table E4, Column (1)). We calculate each elasticity as the ratio of the main reduced-form effect of the supply shock on the outcome to its estimated effect on the log electricity price. At the mean post-phase-out supply shock, the 2010-total-salary-weighted mean predicted log-price change is 8.4 log points, corresponding to an average electricity-price increase of approximately 9.0% in levels. This estimated price response captures only the change in the market price and does not include the shadow price associated with physical rationing.

The structural model yields corresponding price-only elasticities of -0.74, -0.15, -0.15, and -0.17 (Column (2)). We calculate the plant-level responses using the model's closed-form expressions. For example, the own-price elasticity of electricity demand is

$$\epsilon_{E,i} \equiv \frac{\partial \ln E_i}{\partial \ln p^E} = -\sigma + (\sigma - \eta) s^C_{E,ij},$$

where $s^C_{E,ij}$ is electricity's share of the plant's unit cost. We then aggregate the plant-level elasticities using 2010 total-salary weights. Appendix E-3.2 describes the derivation.[17]

7.2 Counterfactual: Regional Market Integration

We use the estimated production framework to evaluate regional market integration as a supply-side intervention that can mitigate the economic costs of electricity supply disruptions. Section 5.6 shows that manufacturing plants have limited capacity to adapt to supply shocks. We therefore focus on a policy that improves electricity-market functioning and lowers electricity prices.

Japan's electricity market has historically been segmented because regulations limited interregional electricity transmission. This segmentation can create dispatch inefficiencies: high-marginal-cost generators may serve local demand while lower-cost capacity elsewhere remains idle (Hausman, 2025). It also limits reserve sharing, requiring regions to maintain redundant backup capacity and raising system-wide costs. The potential gains from greater market integration have therefore attracted substantial policy attention. We quantify the benefits of relaxing these constraints through interregional trade and more efficient reserve sharing.

We quantify the resulting changes in output and producer surplus attributable to electricity supply shocks. In particular, we approximate the policy-induced change in aggregate

[17] Our own-price elasticity estimates are consistent with the quasi-experimental evidence and are closest to von Graevenitz and Rottner (2024), who report estimates of approximately -0.4 to -0.6. Previous studies report industrial own-price elasticities ranging from -0.09 to -0.20 (Gerster and Lamp, 2024) and values as large as -1.5 in settings with substantial price shocks (Martin et al., 2014).

producer surplus as

$$\Delta\Pi \approx \sum_i \Pi_i\left(p^E_{\text{base}}\right)\ (1-\eta)\, s^C_{E,i}\, \Delta \ln p^E, \tag{17}$$

where $s^C_{E,i}$ is plant i's electricity cost share and $\Delta \ln p^E$ is the policy-induced change in the electricity price. This approximation follows from the model's electricity-price elasticities; Appendix E-3 describes the counterfactual simulation procedure.

Estimating these effects requires counterfactual electricity prices under market integration (Appendix F-0.1). We estimate regional supply curves and solve for integrated-grid equilibrium prices, allowing lower-cost generators to serve a wider market and enabling reserve sharing. Full integration reduces the national average log electricity price by 2.8 log points relative to the observed scenario.[18] The mean log-price increase relative to the no-shock baseline falls from 8.4 to 5.7 log points, corresponding to level increases of approximately 9.0% and 5.8%, respectively. Price changes remain heterogeneous, with large declines in high-cost demand centers such as Tokyo and offsetting increases in low-cost exporting regions (Table F1, Column (2)).

Table 6 summarizes the simulation results. Under the price-only structural baseline (the price-contribution row of Panel B), electricity use declines by 6.2%, while labor and materials each decline by 1.3%. Real output and producer surplus fall by 1.5% and 1.2%, respectively. Evaluated at 2010 levels, the output decline corresponds to an annual loss of 4,231 billion yen, compared with the reduced-form estimate of 4,920 billion yen.[19] Producer surplus falls by 677 billion yen.

Market integration with reserve sharing substantially mitigates these costs (Panel C). It reduces the decline in electricity use from 6.2% to 4.2% and the output decline from 1.5% to 1.0%. Annual output losses fall from 4,231 billion yen to 2,864 billion yen, while producer-surplus losses fall from 677 billion yen to 458 billion yen. Both losses decline by approximately 32.3% relative to the baseline.

These results indicate that limited interregional coordination amplified the costs of the post-Fukushima electricity supply shock. Greater market integration recovers nearly one-third of the price-induced losses by improving system-wide dispatch and reserve sharing.

[18] The implied price effects are broadly consistent with existing evidence on market integration. Gonzales et al. (2023) estimate that Chile's 2017 integration reform reduced generation costs by 3–7% in the short run, and by 8–18% when accounting for investment responses. Wolak (2015) finds that relieving transmission congestion in New Zealand lowers wholesale prices by roughly 3% on average. Our estimates are smaller than those in Ryan (2021), who find larger gains in settings with severe transmission shortages.

[19] These correspond to approximately US$35 billion and US$41 billion, respectively, at an exchange rate of 120 yen per dollar.

8 Conclusion

This paper shows that the persistent electricity supply shock following the Fukushima accident imposed sizable costs on Japanese manufacturing. A 6.4% decline in electricity use reduced revenue by 1.8%, implying annual losses of roughly 5.3 trillion yen despite electricity's 2% expenditure share. Plants showed little adjustment through technology, product mix, fuel substitution, or production reallocation. The estimates do not establish whether self-generation mitigated output losses.

Our structural model explains these losses through strong input complementarity and capital idleness and closely matches the reduced-form estimates. The counterfactuals show that higher electricity prices drove most of the output decline and that greater grid integration could recover nearly one-third of the baseline losses. The results highlight the importance of affordable, efficiently allocated electricity for industrial resilience.

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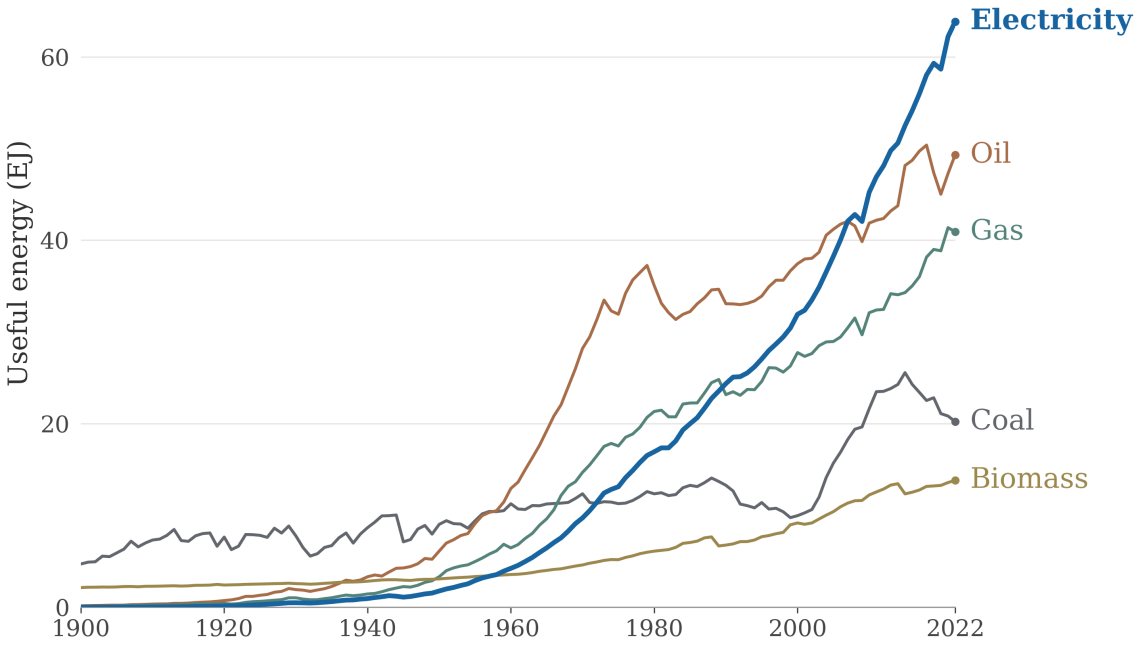
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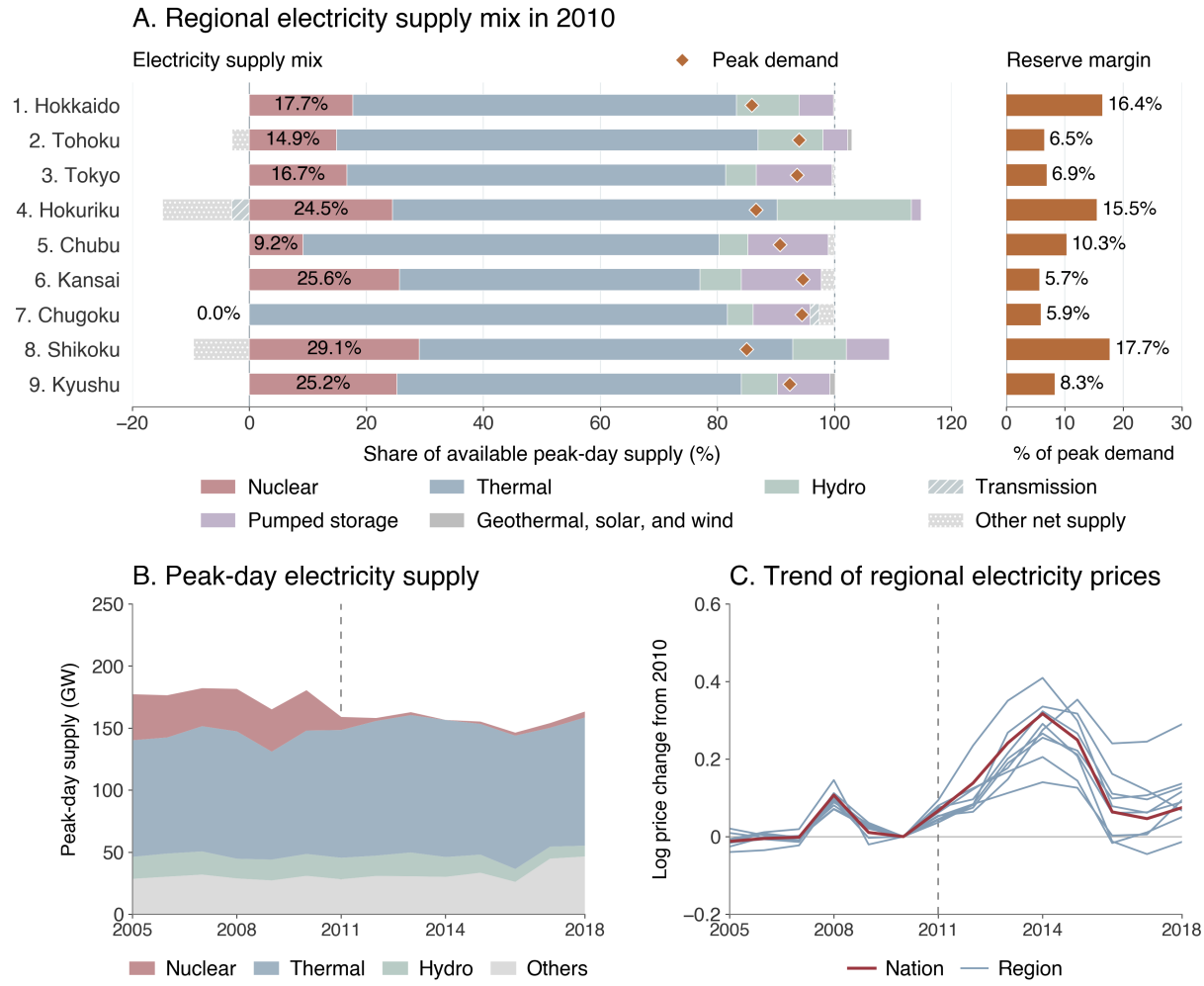
Figures and Tables

Figure 1: Global useful energy by energy carrier, 1900–2022



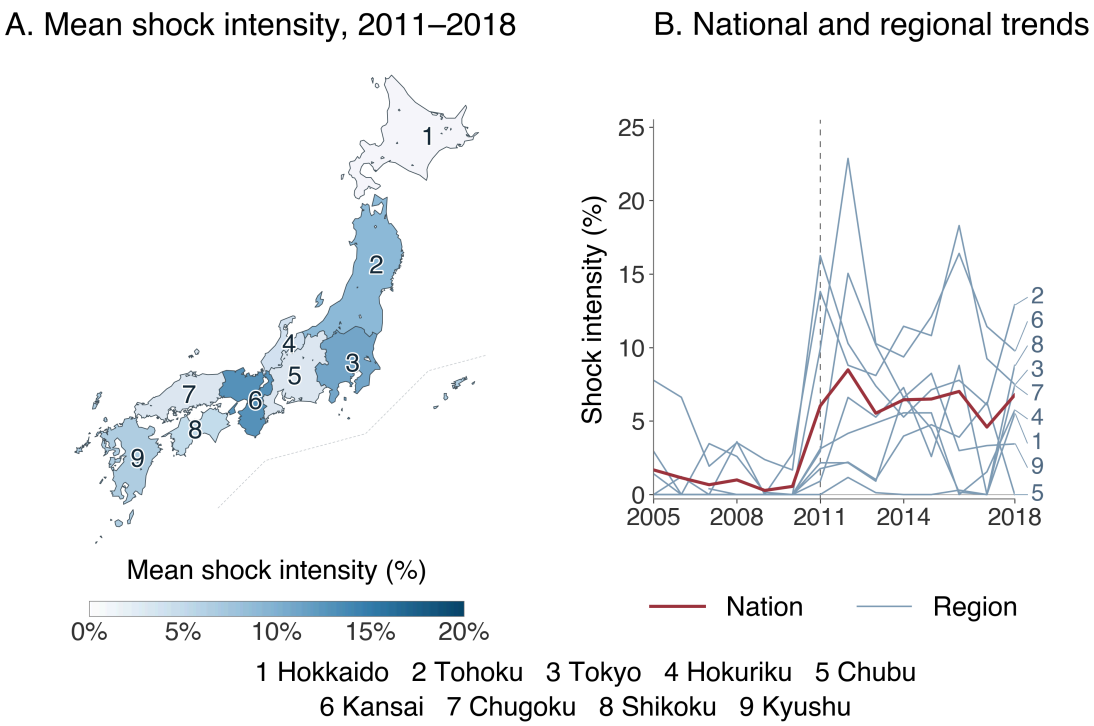
Notes: The figure shows global useful energy supplied by each energy carrier during 1900–2022, measured in exajoules (EJ). Useful energy accounts for losses when energy is converted into end-use services. Non-energy uses, such as feedstocks, are excluded. Source: IIASA and Ember (2025).

Figure 2: Nuclear shutdown and electricity supply shocks



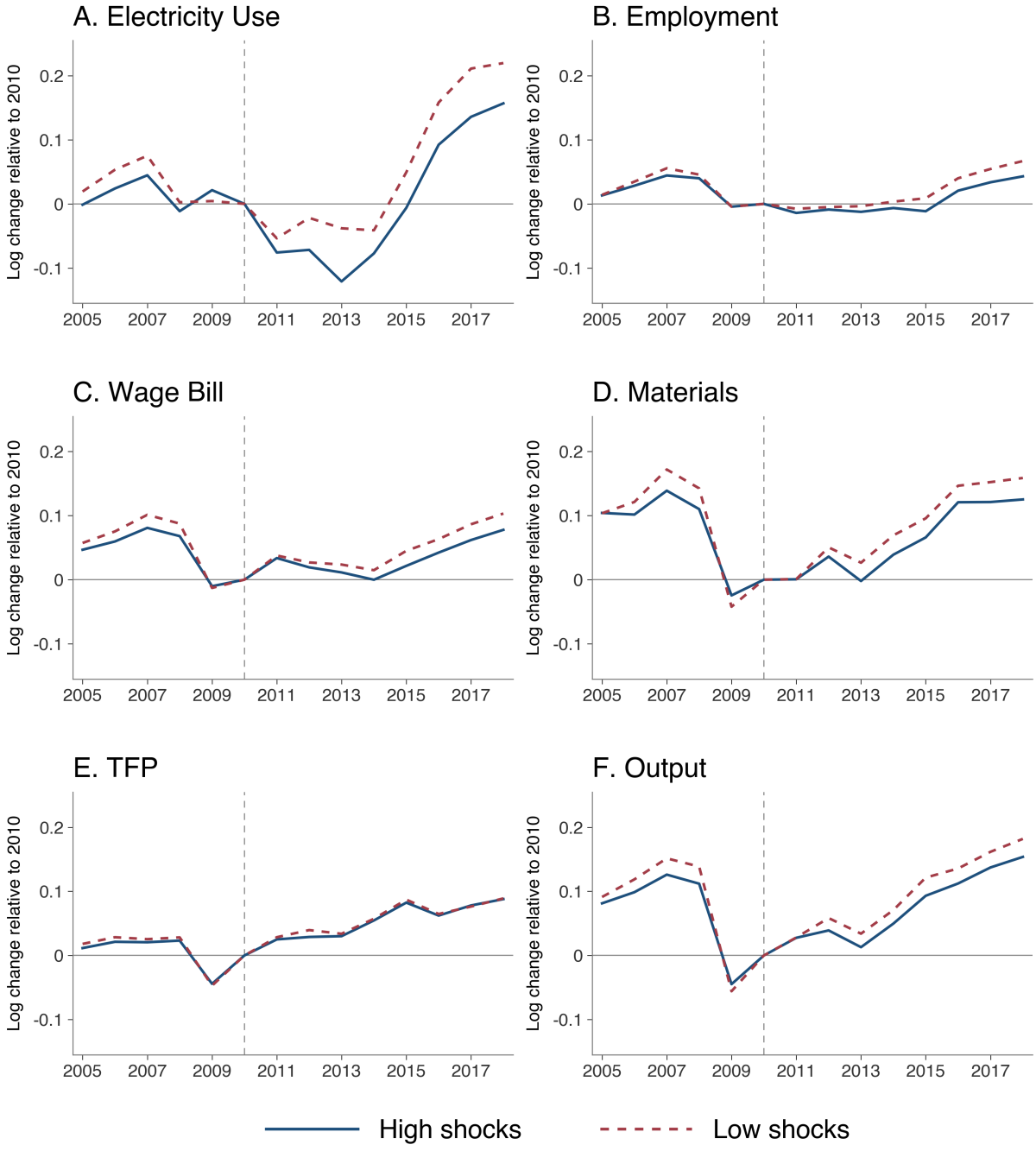
Notes: Panel A shows each source's share of available peak-day electricity supply and reserve margins in 2010 for the nine mainland regions. Reserve margins equal available supply minus peak demand, divided by peak demand. Transmission denotes interutility exchanges. Other net supply includes electricity received from and delivered to non-utility suppliers and the wholesale market. Negative values indicate net deliveries; these need not cross regional boundaries. Panel B shows peak-day electricity supply by source in GW. Others includes purchases, other sources, and net adjustments. Panel C shows log electricity-price changes relative to 2010. The bold line represents the national trend, and thin lines represent individual regions. In Panels B and C, the vertical dashed line marks 2011.

Figure 3: Distribution and trend of electricity supply shocks



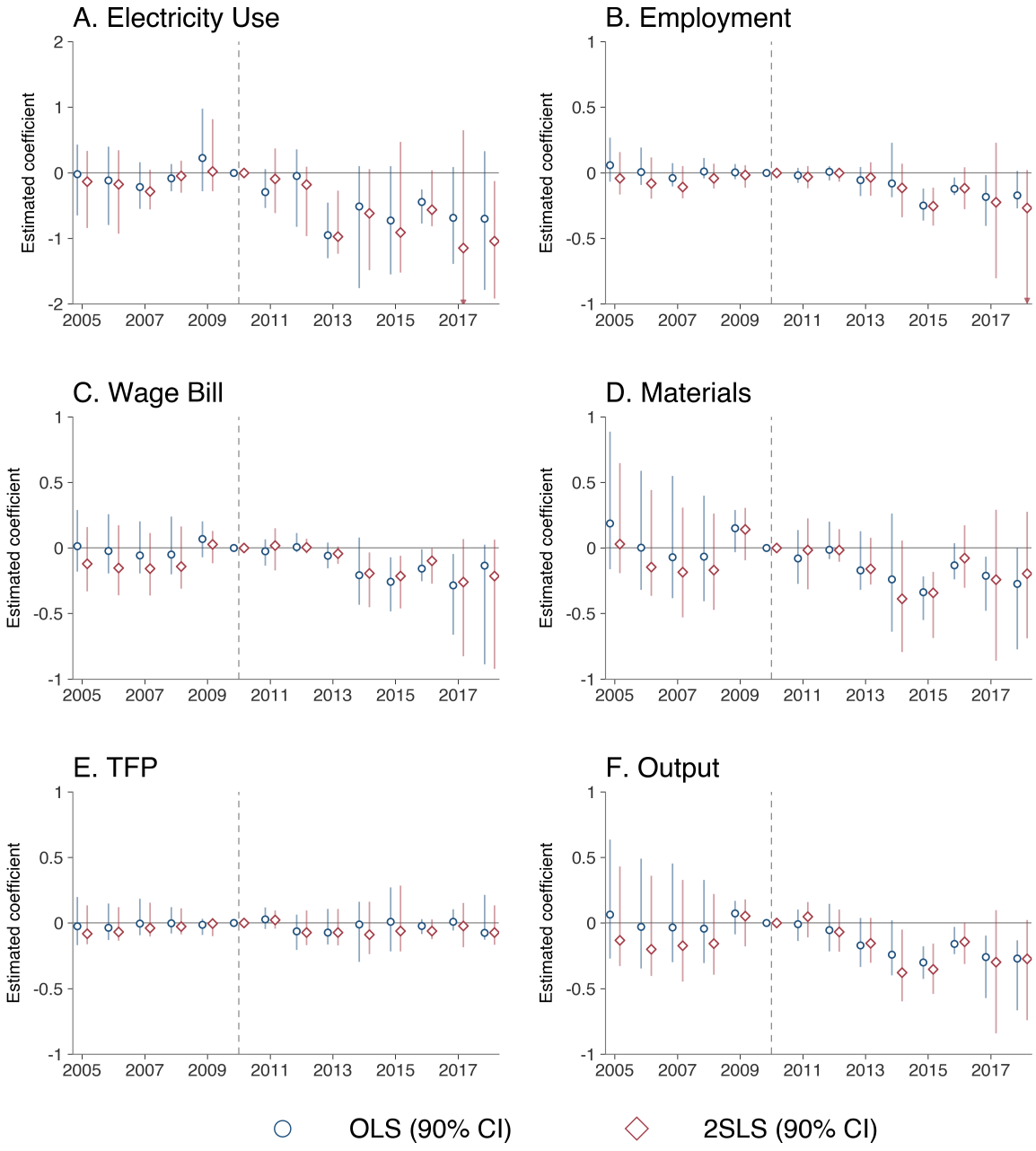
Notes: Panel A shows mean shock intensity in 2011–2018 for each electricity service area. Panel B shows annual shocks in 2005–2018. The bold red line is the national average, weighted by the log of 2010 peak demand; thin blue lines represent individual regions. The dashed line marks 2011. Shocks are expressed as percentages of 2010 peak demand; see equation (1) for the definition.

Figure 4: Trends of various measures of manufacturing performances between high-shock and low-shock regions



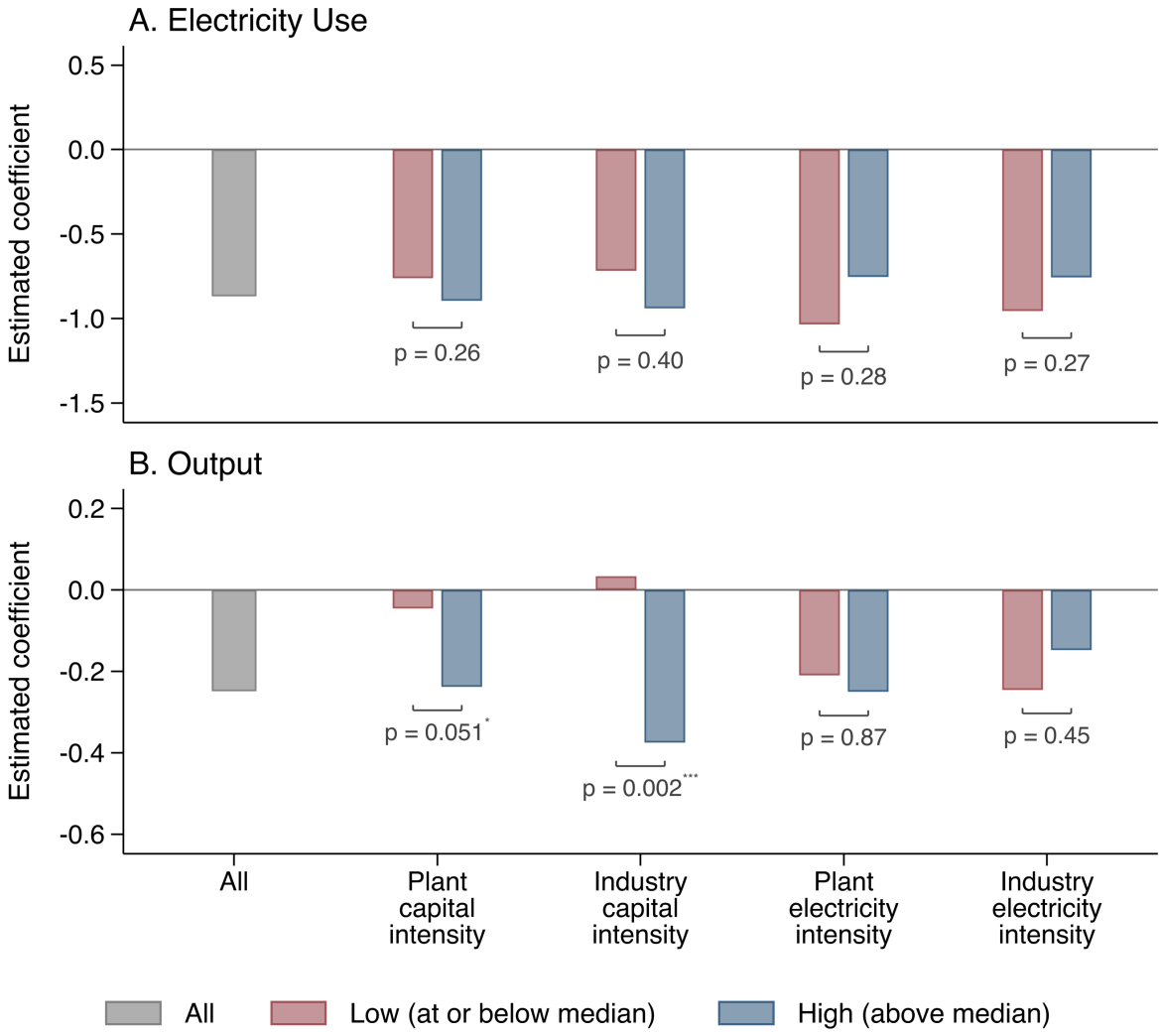
Notes: We classify regions with mean supply shocks during 2011–2018 at or above the median across the nine regions as high-shock regions and the remaining regions as low-shock regions. Outcomes are normalized by their 2010 values, so the y-axis represents changes relative to 2010.

Figure 5: Event study results of effects of supply shock on various manufacture's activities



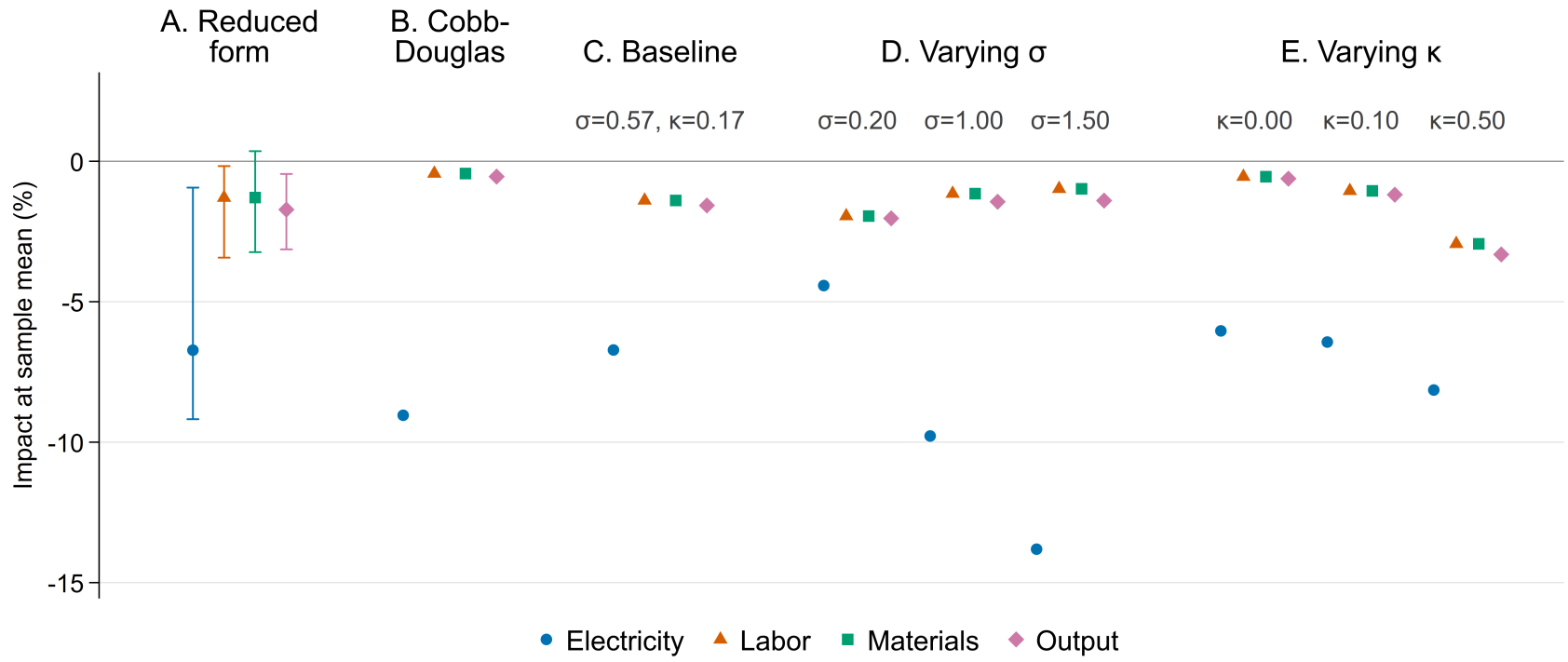
Notes: The figure reports event-study estimates from equation (6). Blue circles denote OLS estimates, and red diamonds denote 2SLS estimates. Each point and its 90% confidence interval come from a separate regression using establishment-level data for the indicated year, with outcomes measured as log changes relative to 2010. The 2010 points are normalized to zero. Vertical bars show 90% confidence intervals obtained using a wild cluster bootstrap with clustering by region. Confidence intervals extending beyond the plotted ranges are truncated at the axis limits (-2 to 2 in Panel A and -1 to 1 in Panels B–F). For pre-accident years, we use each region's mean supply shock over 2011–2018. All regressions include three-digit-industry-by-year fixed effects.

Figure 6: Heterogeneous impacts of electricity supply shocks



Notes: The figure reports 2SLS coefficients from equation (5) for electricity use (Panel A) and output (Panel B). Gray bars show the full-sample estimates. Red and blue bars show estimates for establishments at or below and above the median, respectively, of the indicated establishment- or three-digit-industry characteristic measured in 2010. Labels below the brackets report wild-cluster-bootstrap p-values for tests of equality between the two subgroup coefficients, clustering by region. Stars use unrounded p-values: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All regressions include year-by-three-digit-industry fixed effects.

Figure 7: Reduced-Form Estimates and Structural Simulations



Notes: The figure compares reduced-form impacts at the weighted mean post-phase-out supply shock with structural predictions. Reduced-form whiskers show 90% confidence intervals. Panel B reports the Cobb–Douglas benchmark, and Panel C reports the calibrated baseline with $\sigma = 0.57$ and $\kappa = 0.17$. Panel D varies $\sigma \in \{0.20, 1.00, 1.50\}$ while holding $\kappa = 0.17$ and baseline cost shares fixed. Panel E varies $\kappa \in \{0.00, 0.10, 0.50\}$ while holding $\sigma = 0.57$. Within every specification, markers appear in the order electricity, labor, materials, and output. Generator owners retain their simulated grid-electricity response, while their structural non-electricity and output responses are set to zero.

Table 1: Estimated Output Elasticities

	Structural Estimates			Benchmark
	Mean	p25	p75	OLS
	(1)	(2)	(3)	(4)
Materials (θ_M)	0.439	0.387	0.498	0.485
Electricity (θ_E)	0.015	0.008	0.018	0.066
Labor (θ_L)	0.358	0.269	0.433	0.431
Capital (θ_K)	0.144	0.088	0.174	0.042

Notes: The table summarizes raw output elasticities estimated over 2005-2010 using the Gandhi, Navarro, and Rivers (2020) method. Columns (1)-(3) report the mean and the 25th and 75th percentiles across 128 industries. Column (4) reports the mean OLS benchmark across the same industries, from regressions of log deflated output on log inputs.

Table 2: First-Stage Regression: Impact of Supply Composition on Supply Shocks

	Dep. Var.: Supply shock	
	2011–2012	2013–2018
	(1)	(2)
Predicted nuclear supply growth	-0.58***	-0.26
	(0.05)	(0.06)
Reserve margin in 2010	-1.39***	-0.83***
	(0.08)	(0.12)
Industry-by-year FE	Yes	Yes
Region clusters	9	9
Obs.	268,264	383,863
Joint p-value	< 0.001***	0.016**
F-statistics	173	28

Notes: We use establishment-year data, consistent with the second stage. The dependent variable is the regional electricity supply shock. Nuclear exposure is 2010 available nuclear capacity divided by 2010 peak demand, interacted with national nuclear-supply growth relative to 2010. The reserve margin is 2010 available supply less peak demand, divided by peak demand. Column (1) uses data from 2005-2010 and 2011-2012; Column (2) uses data from 2005-2010 and 2013-2018. Treatment and instruments are set to zero before 2011. All regressions include three-digit-industry-by-year fixed effects. Parentheses report standard errors clustered by electricity region. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3: Effects of Supply Shocks on Inputs and Outputs

	Dep Var: $\Delta \ln(Y) = \ln(Y) - \ln(Y_{2010})$					
	Electricity use	Employ-ment	Wage bill	Materials	TFP	Output
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. OLS						
Supply shock: 2011–2012	-0.14*	0.00	0.00	-0.04**	-0.03	-0.04
	(0.10)	(0.01)	(0.02)	(0.02)	(0.03)	(0.02)
Supply shock: 2013–2018	-0.60**	-0.13**	-0.17**	-0.19**	-0.03	-0.21***
	(0.15)	(0.04)	(0.07)	(0.06)	(0.04)	(0.04)
Panel B. 2SLS						
Supply shock: 2011–2012	-0.15	-0.01	0.01	-0.01	-0.05	-0.03
	(0.18)	(0.01)	(0.03)	(0.04)	(0.04)	(0.03)
Supply shock: 2013–2018	-0.87*	-0.15*	-0.15*	-0.22	-0.07	-0.25**
	(0.21)	(0.07)	(0.09)	(0.10)	(0.06)	(0.07)
Industry-by-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region clusters	9	9	9	9	9	9
Obs.: 2011–2012	268,264	268,264	268,264	268,264	260,147	268,264
Obs.: 2013–2018	383,863	383,863	383,863	383,863	371,726	383,863

Notes: Outcomes are log changes from 2010. All regressions include three-digit-industry-by-year fixed effects. Phase-out estimates use 2005-2010 and 2011-2012; post-phase-out estimates use 2005-2010 and 2013-2018. Treatment and both instruments are set to zero through 2010. The 2SLS instruments are 2010 nuclear capacity divided by peak demand times common national nuclear-supply growth, and the 2010 reserve margin. Parentheses report standard errors clustered by electricity region. Stars use wild-cluster-bootstrap Wald p-values with Rademacher weights. Observation counts include the pre-period. Stars use unrounded p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 4: Effects of Supply Shocks on Establishment Behavior

	Product-level		Outsourcing Share in Materials	Working Hours		
	Switching (1)	Intensity (2)	(3)	Total (4)	Scheduled (5)	Overtime (6)
Panel A. OLS						
Supply shock: 2011–2012	0.02	0.02	-0.01	0.01	0.00	0.05
	(0.06)	(0.02)	(0.01)	(0.02)	(0.01)	(0.18)
Supply shock: 2013–2018	-0.02	0.04	-0.01	-0.08	-0.06*	-0.24
	(0.06)	(0.02)	(0.01)	(0.04)	(0.02)	(0.24)
Panel B. 2SLS						
Supply shock: 2011–2012	0.04	0.02	-0.01	0.02	0.01	0.04
	(0.06)	(0.02)	(0.01)	(0.03)	(0.01)	(0.22)
Supply shock: 2013–2018	-0.02	0.04	-0.01	-0.10*	-0.09*	-0.15
	(0.10)	(0.03)	(0.01)	(0.05)	(0.03)	(0.38)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Region clusters	9	9	9	9	9	9
Obs.: 2011–2012	268,264	168,198	265,106	365	365	365
Obs.: 2013–2018	383,863	229,729	378,764	552	552	552

Notes: Column (1) indicates product switching. Column (2) is the log change in product-level electricity intensity. Column (3) is the level change in outsourcing as a share of materials. Columns (4)–(6) are log changes in average monthly total, scheduled, and overtime hours per worker. All changes are relative to 2010. Columns (1)–(3) use unweighted establishment-level regressions with three-digit-industry-by-year fixed effects. Columns (4)–(6) use prefecture-level data, include year fixed effects, and use fixed 2010 CM establishment counts as regression weights. Phase-out estimates use 2005–2010 and 2011–2012; post-phase-out estimates use 2005–2010 and 2013–2018. Parentheses report standard errors clustered by electricity region, using the dominant region for prefectures. Significance levels use wild cluster bootstrap p-values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5: Effects of Supply Shocks on Energy Usage

	Fuel and steam for:			
	Production	Self-generation	Electricity generated	Ownership
	(1)	(2)	(3)	(4)
Panel A. OLS				
Supply shock: 2011–2012	-0.04	0.16	0.62	0.022***
	(0.08)	(0.47)	(0.83)	(0.005)
Supply shock: 2013–2018	-0.16	2.42	2.89*	-0.002
	(0.15)	(0.97)	(0.72)	(0.024)
Panel B. 2SLS				
Supply shock: 2011–2012	-0.09	0.36	1.11	0.031***
	(0.07)	(0.68)	(1.05)	(0.006)
Supply shock: 2013–2018	-0.24	1.28	3.82*	0.011
	(0.28)	(1.23)	(1.38)	(0.047)
Industry-by-year FE	Yes	Yes	Yes	Yes
Region clusters	9	9	9	9
Obs.: 2011–2012	98,742	3,696	4,779	148,450
Obs.: 2013–2018	134,908	5,123	7,251	206,779

Notes: The data combine the Energy Consumption Survey and Current Survey of Petroleum and Other Energy Consumption. Columns (1)–(3) use log changes from 2010; Column (4) uses the level change in generator ownership. Columns (2)–(3) are restricted to observations classified as owners. Regressions are unweighted and include two-digit-industry-by-year fixed effects. Phase-out estimates use 2007–2010 and 2011–2012; post-phase-out estimates use 2007–2010 and 2013–2018. Parentheses report standard errors clustered by electricity region. Significance levels use wild cluster bootstrap p-values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6: Structural Simulation Impacts and Counterfactuals

	Post-phase-out impacts				Monetary losses (billion yen)	
	Electricity (1)	Labor (2)	Materials (3)	Output (4)	Output (5)	Profit (6)
Panel A. Reduced-form impacts at the mean shock						
Reduced-form estimate	-6.6%	-1.3%	-1.3%	-1.7%	-4,920	–
Panel B. Structural estimation						
Total impact	-6.7%	-1.4%	-1.4%	-1.6%	-4,546	-727
Price contribution	-6.2%	-1.3%	-1.3%	-1.5%	-4,231	-677
(Share of total impact)	(92.7%)	(93.0%)	(93.0%)	(93.0%)	(93.0%)	(93.0%)
Rationing contribution	-0.5%	-0.1%	-0.1%	-0.1%	-316	-50
(Share of total impact)	(7.3%)	(7.0%)	(7.0%)	(7.0%)	(7.0%)	(7.0%)
Panel C. Counterfactual analysis						
Market integration	-4.2%	-0.9%	-0.9%	-1.0%	-2,864	-458

Notes: Columns (1)–(4) report annualized log impacts during the post-phase-out period, expressed in percent. Columns (5) and (6) report annual aggregate output and profit losses in billion constant 2010 yen. Panel A reports the main reduced-form IV estimates multiplied by the mean shock. Panel B reports predictions from the preferred dynamic-labor structural model. The total impact compares predicted electricity prices and seasonal rationing with the 2010 baseline. The price contribution changes predicted prices while holding rationing at the no-rationing baseline. The rationing contribution is the difference between the total and price-only impacts and therefore includes their interaction. Panel C compares the price-only baseline reported in the price-contribution row of Panel B with full market integration and reserve sharing. Percentage responses are weighted by 2010 total salary. Aggregate losses apply these weighted responses to aggregate 2010 output. Contribution shares are calculated from underlying unrounded estimates. Generator owners retain their simulated grid-electricity response, while their labor, materials, output, and profit responses are set to zero based on the heterogeneous IV evidence.

ONLINE APPENDIX

Electricity Supply Shocks and Manufacturing: Evidence from the Fukushima Nuclear Accident

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A Data Appendix

A-1 Census of Manufactures

A-1.1 Sampling

We use establishment-level data from Japan's Census of Manufactures (CMF) for 2005–2018. The CMF surveys manufacturing establishments with at least four employees. It administers a simplified questionnaire to establishments with 4–29 employees and a detailed questionnaire to those with 30 or more employees. We use the detailed questionnaire because it reports the output, input, inventory, and fixed-asset information required for our analysis.

In 2011 and 2015, the Economic Census for Business Activity collected the corresponding manufacturing data in place of the CMF (METI, 2012; Ministry of Internal Affairs and Communications and Ministry of Economy, Trade and Industry, 2012, 2016). The Ministry of Internal Affairs and Communications and METI jointly conducted the Economic Census. We use the official longitudinal identifier to link the 2011 records to the CMF. For 2015, we link records across years using normalized establishment names within prefecture–municipality cells.

We assign each establishment to the service territory of one of Japan's ten regional electricity utilities. The prefecture identifies the utility for most establishments. For establishments near utility boundaries that cross prefectural borders, we use municipality and postal codes. We exclude Okinawa because its isolated electricity system does not provide variation comparable to that of the nine interconnected utility regions.

A-1.2 Variable Construction

We measure output using total revenue and inputs using expenditures, with all values adjusted for taxes and deflated. The tax adjustment accounts for Japan's consumption tax increase from 5% to 8% in April 2014.

Revenue. We measure nominal output as total receipts adjusted for changes in inventories of finished and semi-finished goods. We deflate nominal output using industry-specific output price indexes from the Bank of Japan (2026).

Electricity. We deflate electricity expenditure using average electricity prices. We obtain average electricity prices from FEPC (2026). The resulting measure is expressed in 2010-price expenditure units rather than physical units such as kilowatt-hours.

Materials. We measure nominal materials as total intermediate-input expenditure, adjusted for changes in inventories of raw materials and fuels. We deflate materials using industry-specific input price indexes from the Bank of Japan (2026).

Labor. Our primary measure of labor input is total cash wages and salaries, deflated using the Consumer Price Index from the Statistics Bureau of Japan (2026). We also use the number of employees. We divide employment into regular and non-regular workers. Non-regular workers include part-time workers and workers dispatched or seconded from other firms.

Capital. We measure installed capital using the beginning-of-year book value of tangible fixed assets, deflated using capital-stock deflators constructed from Cabinet Office (2026) data. We calculate capital rental costs by multiplying deflated capital stock by an industry-year rental rate, measured as nominal capital cost divided by real net capital stock in the JIP database (Cabinet Office, 2023).

A-1.3 Additional Measures

Multi-establishment firms. We use firm identifiers in the CMF to examine whether firms reallocate activity across establishments in response to localized electricity supply shocks (Section 5.6). To examine production reallocation among existing establishments, we follow a fixed set of establishments belonging to each firm in 2010. For each firm-region-year, we aggregate electricity use and output across the firm's establishments outside the focal region. We then calculate the log change in each outside-region total relative to the corresponding 2010 total. For example, when Tokyo is the focal region, we compare a firm's electricity use and output at all of its non-Tokyo establishments with their 2010 levels.

Product switching. We also use product-level CMF records to measure product switching and changes in product mix (Section 5.6). We define the switching indicator as one if the establishment's set of reported products differs from its 2010 set because it added or dropped at least one product. To construct product-level electricity intensity, we first calculate the electricity-use-to-sales ratio for each single-product establishment in 2010 and take the median ratio within each product. We then assign these product-specific intensities to each establishment's reported products and calculate their sales-weighted average in each year.

Entry and exit. We use publicly available aggregate establishment-count data from the 2009, 2012, 2014, and 2016 Economic Census waves to examine establishment turnover. We rely on these aggregate data because our restricted-access microdata cover only establishments with 30 or more employees. The aggregate data report the numbers of establishments, openings, and closures by prefecture, industry, and survey wave, which we map to electricity regions. The data do not distinguish establishment entry and exit from relocation across regions.

A-2 Energy Consumption Statistics (ECS)

We combine the ECS with the Current Survey of Petroleum and Other Energy Consumption (CSPOEC), which covers nine energy-intensive manufacturing industries. The ECS excludes establishments covered by CSPOEC, so the two surveys do not overlap. Together, the surveys target all establishments on the self-generation roster. Compared with the CMF, the pooled sample is more concentrated among large, energy-intensive establishments and those with on-site generation.

We construct total fuel consumption in common energy units in two steps since the surveys report a broad range of energy inputs. First, we standardize each fuel's reported quantity because establishments report either physical quantities or monetary values and may use different units. Second, we convert the resulting quantities into a common energy unit using conversion factors based primarily on the 2013 standard gross calorific values published by the ANRE (Agency for Natural Resources and Energy, 2022). We aggregate energy content across fuel types to express total fuel use in gigajoules (GJ).

We construct fuel consumption for on-site generation using the reported purpose of fuel consumption. We define fuel used for self-generation as fuel consumed by boilers used for electricity generation, cogeneration systems, and diesel generators. For each fuel type, we calculate fuel used directly in production as total fuel consumption minus fuel used by these self-generation facilities.

The ECS does not directly report equipment ownership, so we infer self-generator ownership from observed self-generation. We classify an establishment as owning a self-generator beginning in the first year self-generation reaches at least 10% of total electricity consumption; ownership persists thereafter. For the heterogeneity analysis by pre-accident ownership, establishments are classified as owners if this share reaches at least 10% in any year during 2007–2010.

A-3 Additional Datasets

A-3.1 Robustness-Control Variables

This appendix describes the regional variables used in the robustness checks in Section 5.5. We construct supply-chain exposure measures using prefecture-by-industry input-output tables from RIETI (RIETI, 2005). We define disaster-affected areas as Iwate, Miyagi, and Fukushima prefectures, which accounted for more than 99% of deaths and missing persons from the earthquake and tsunami. The direct exposure measure captures the share of a prefecture-industry's intermediate inputs sourced from disaster-affected areas. For example, if an iron factory sources 50% of its materials from disaster-affected areas, its direct exposure is 50%. The indirect exposure measure incorporates upstream input-output linkages to

capture exposure through suppliers' suppliers. For example, a machinery establishment that procures iron from suppliers using materials from disaster-affected areas would have indirect exposure. We use the 2005 tables, the latest available wave before the Fukushima accident. Because these exposure measures are time-invariant, the robustness specifications interact the pre-disaster measures with year indicators.

We measure local demand conditions using indicators of economic activity, population, spending, employment, and consumer prices. Gross prefectural product in the primary and tertiary sectors, total and working-age populations, planned residential construction expenditure, and household expenditure come from prefecture-level statistics published by the Ministry of Internal Affairs and Communications (MIC, 2025). We measure employment outside manufacturing using the R-JIP2021 database. To capture local price conditions, we use the consumer price index excluding food (except alcohol) and energy for each prefectural capital, or Tokyo's 23 wards. All controls enter as log changes from 2010.

A-3.2 Working Hours

We obtain prefecture-level manufacturing working-hours data from the Ministry of Health, Labour and Welfare's Monthly Labour Survey (MHLW, 2005–2018). We use observations for establishments with at least 30 regular employees. The survey reports average monthly scheduled and overtime hours per worker; total working hours equal their sum. We construct annual averages using monthly regular-worker counts as weights and require observations for all 12 months. For each measure, we calculate log changes relative to its 2010 level.

Table A1: Summary Statistics

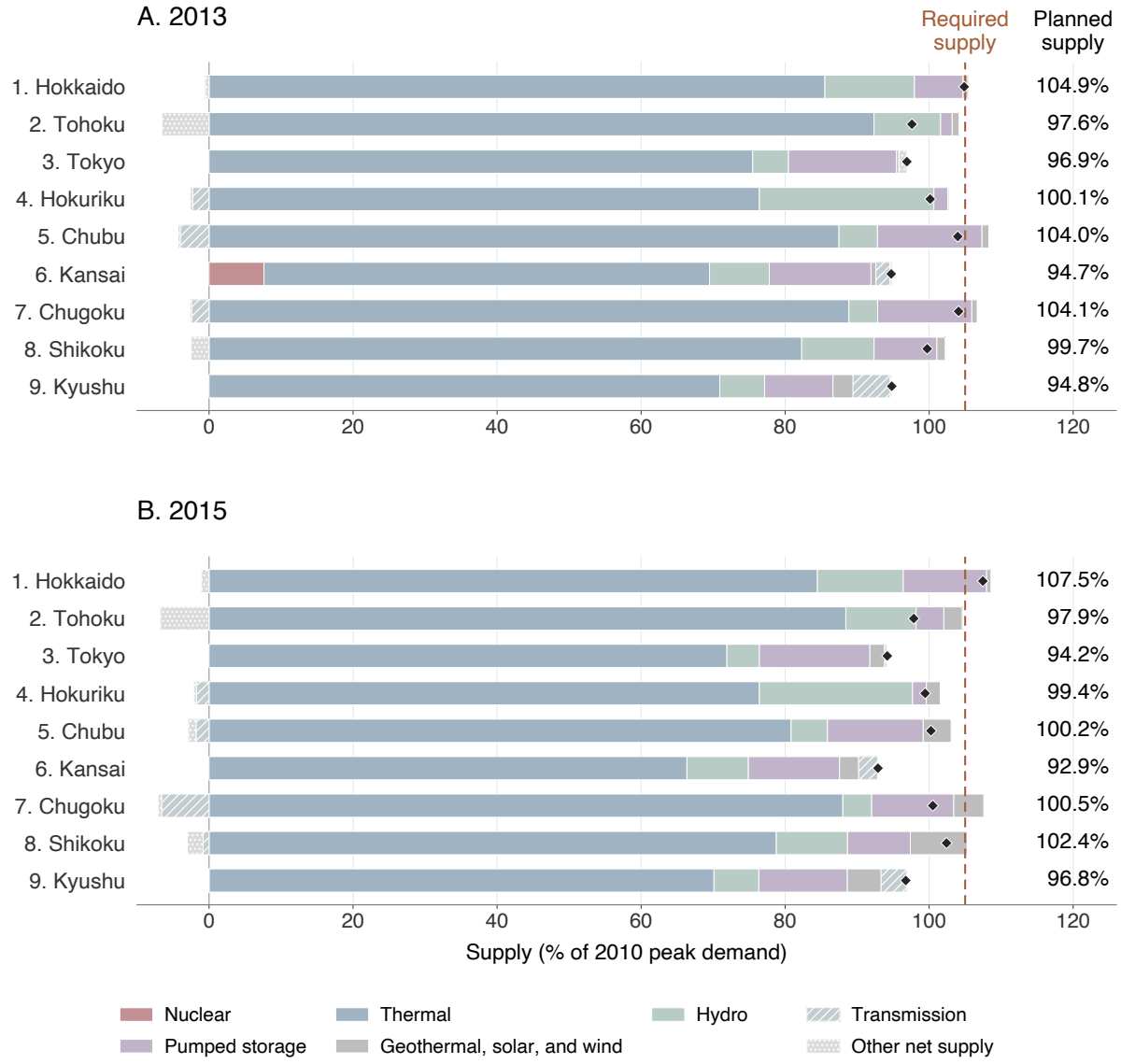
	Mean (1)	S.D. (2)
Panel A. Establishment Characteristics		
Revenue (million yen)	5,122	11,647
Employment (headcount)	128	168
Capital stock (million yen)	1,058	2,543
Multi-establishment firm (yes = 1)	0.48	0.50
Panel B. Revenue Shares		
Labor	0.22	0.14
Capital	0.033	0.042
Materials	0.51	0.20
Electricity	0.020	0.024
Panel C. Establishment Energy Usage		
Fuel input for production (TJ)	15.2	38.5
Fuel input for self-generation (TJ)	0.9	6.6
Self-generated electricity (TJ)	2.2	15.4
Grid electricity purchased (TJ)	16.4	40.2
Generator ownership (10% threshold)	0.06	0.23

Notes: Panels A and B use 448,148 establishment-year observations from the CMF in 2005–2018. Monetary values are in 2010 prices. Panel B reports nominal input expenditure divided by nominal revenue. Panel C draws on 373,218 records from the ECS in 2007–2018. All panels cover the nine mainland electricity regions.

B Empirical Strategy

B-1 Measuring Electricity Supply Shocks

Figure B1: Planned regional electricity supply by source in 2013 and 2015



Notes: Panels A and B show planned peak-day electricity supply by source in 2013 and 2015 for the nine mainland regions. Values are percentages of each region's 2010 peak demand. Diamonds show total planned supply. The dashed line marks required supply: 2010 peak demand plus a 5% reserve margin. Transmission denotes interutility exchanges. Other net supply includes electricity received from and delivered to non-utility suppliers and the wholesale market. Negative values indicate net deliveries; these need not cross regional boundaries.

B-2 Derivation of the Small-Share Benchmark

Suppose P denotes the output price and P^K, P^L, P^M, and P^E denote input prices. An establishment solves

$$\max_{K,L,M,E} P\Omega F(K,L,M,E) - P^K K - P^L L - P^M M - P^E E. \tag{B1}$$

Taking logs gives

$$\ln Y = \ln \Omega + \ln F(K,L,M,E). \tag{B2}$$

Differentiating with respect to shock severity $\mathcal{S}$ yields

$$\frac{d\ln Y}{d\mathcal{S}} = \frac{d\ln \Omega}{d\mathcal{S}} + \frac{1}{F}\frac{dF}{d\mathcal{S}}. \tag{B3}$$

By the chain rule,

$$\begin{aligned}\frac{1}{F}\frac{dF}{d\mathcal{S}} &= \frac{F_K}{F}\frac{dK}{d\mathcal{S}} + \frac{F_L}{F}\frac{dL}{d\mathcal{S}} + \frac{F_M}{F}\frac{dM}{d\mathcal{S}} + \frac{F_E}{F}\frac{dE}{d\mathcal{S}} \\ &= \frac{F_K K}{F}\frac{d\ln K}{d\mathcal{S}} + \frac{F_L L}{F}\frac{d\ln L}{d\mathcal{S}} + \frac{F_M M}{F}\frac{d\ln M}{d\mathcal{S}} + \frac{F_E E}{F}\frac{d\ln E}{d\mathcal{S}}.\end{aligned} \tag{B4}$$

Define the output elasticity with respect to each input as

$$\theta_K \equiv \frac{F_K K}{F}, \qquad \theta_L \equiv \frac{F_L L}{F}, \qquad \theta_M \equiv \frac{F_M M}{F}, \qquad \theta_E \equiv \frac{F_E E}{F}. \tag{B5}$$

Substituting these expressions into the derivative above yields

$$\frac{d\ln Y}{d\mathcal{S}} = \theta_E \frac{d\ln E}{d\mathcal{S}} + \theta_M \frac{d\ln M}{d\mathcal{S}} + \theta_L \frac{d\ln L}{d\mathcal{S}} + \theta_K \frac{d\ln K}{d\mathcal{S}} + \frac{d\ln \Omega}{d\mathcal{S}}. \tag{B6}$$

Since electricity is a flexible input, under price-taking behavior, the first-order condition for electricity is

$$P\Omega F_E = P^E. \tag{B7}$$

Multiplying this condition by electricity use and dividing by nominal revenue $PY = P\Omega F$ gives

$$\theta_E = \frac{F_E E}{F} = \frac{P^E E}{PY} = s_E. \tag{B8}$$

For the small-share benchmark, evaluate this condition at the pre-shock optimum, denoted by stars. Holding labor, materials, capital, and productivity fixed at L^*, M^*, K^*, and Ω^*, respectively, sets the non-electricity terms in equation (3) to zero. The direct first-order effect of the shock is therefore

$$\left.\frac{d\ln Y}{d\mathcal{S}}\right|_{L=L^*,\, M=M^*,\, K=K^*,\, \Omega=\Omega^*} \approx \theta_E^* \frac{d\ln E}{d\mathcal{S}} = s_E^* \frac{d\ln E}{d\mathcal{S}}. \tag{B9}$$

The approximation is local and abstracts from induced adjustments in non-electricity inputs and productivity. This establishes equation (4).

B-3 Production Function Estimation

B-3.1 Estimating Output Elasticities and Productivity

Flexible Inputs In the first stage, we estimate output elasticities of the flexible inputs: $\theta_E \equiv \partial f/\partial e$ (electricity) and $\theta_M \equiv \partial f/\partial m$ (materials). Under the assumption that establishments are price takers in both input and output markets, the first-order condition for a flexible input implies that its log revenue share equals its log output elasticity minus a mean-zero idiosyncratic shock. (See Appendix B-3.3 for the derivation.) For electricity,

$$\ln s_{E,it} = \ln \theta_E - \epsilon_{it}, \tag{B10}$$

where $s_{E,it} \equiv \frac{P_t^E E_{it}}{P_{it} Y_{it}}$ and $P_{it}Y_{it}$ is nominal revenue. Equation (B10) implies that θ_E is identified by the conditional mean of the log electricity expenditure share.[1] We obtain θ_M analogously using the materials share.

Quasi-fixed Inputs In the second stage, we estimate the output elasticities of the quasi-fixed inputs, labor and capital. Using the first-stage estimates of the flexible-input output elasticities, $\widehat{\theta}_E$ and $\widehat{\theta}_M$, we construct the following partial residual for each establishment-year observation:

$$\widehat{\mathcal{Y}}_{it} \equiv y_{it} - \widehat{\theta}_E e_{it} - \widehat{\theta}_M m_{it}. \tag{B11}$$

This partial residual represents log output net of the estimated contributions of electricity and materials. For candidate values of (θ_L, θ_K), we define implied productivity as

$$\widehat{\omega}_{it}(\theta_L, \theta_K) \equiv \widehat{\mathcal{Y}}_{it} - \theta_L l_{it} - \theta_K k_{it}. \tag{B12}$$

For each candidate pair (θ_L, θ_K), we use the implied productivity series to estimate the conditional expectation function $h(\cdot)$ and construct the corresponding productivity innovation as $\widehat{\eta}_{it} = \widehat{\omega}_{it} - \widehat{h}(\widehat{\omega}_{i,t-1})$. We then estimate θ_L and θ_K by setting the sample analogues of $\mathbb{E}[\eta_{it} l_{it}] = 0$ and $\mathbb{E}[\eta_{it} k_{it}] = 0$ equal to zero. Because there are two parameters and two moment conditions, the GMM system is exactly identified.

Total Factor Productivity We use $\widehat{\omega}_{it}$ as our measure of log total factor productivity in the reduced-form analysis.

[1] We impose the normalization $\mathbb{E}[\exp(\epsilon_{it}) \mid k_{it}, l_{it}, m_{it}, e_{it}] = 1$, so that $\mathbb{E}[\ln s_{E,it} \mid k_{it}, l_{it}, m_{it}, e_{it}] = \ln \theta_E$.

B-3.2 Recovering CES Distribution Parameters

For each industry, we first normalize the estimated output elasticities to obtain input cost shares:

$$\hat{s}_j = \frac{\hat{\theta}_j}{\sum_{k\in\{K,L,M,E\}} \hat{\theta}_k}, \qquad j \in \{K, L, M, E\}. \tag{B13}$$

Here, $\hat{\theta}_j$ denotes the estimated output elasticity of input j.

Under cost minimization, these normalized output elasticities correspond to input cost shares. Conditional on a candidate elasticity of substitution σ, we recover the CES distribution parameters as

$$\hat{\alpha}_j(\sigma) = \frac{\hat{s}_j^{1/\sigma}}{\sum_{k\in\{K,L,M,E\}} \hat{s}_k^{1/\sigma}}, \qquad j \in \{K, L, M, E\}. \tag{B14}$$

This parameterization satisfies $\sum_j \hat{\alpha}_j(\sigma) = 1$ and reproduces the estimated cost shares. Appendix B-3.4 provides the detailed derivation.

Taking the ratio of the recovered distribution parameters for any two inputs j and k gives

$$\frac{\hat{\alpha}_j(\sigma)}{\hat{\alpha}_k(\sigma)} = \left(\frac{\hat{s}_j}{\hat{s}_k}\right)^{1/\sigma}. \tag{B15}$$

Thus, inputs with larger estimated cost shares receive larger distribution parameters. Holding the estimated cost shares fixed, a smaller σ implies a more dispersed distribution of these parameters. When substitutability is limited, cost shares are less responsive to differences in the distribution parameters, so larger parameter differences are required to reproduce the observed cost-share differences. Conversely, when σ is large and inputs are more substitutable, smaller parameter differences are sufficient.

B-3.3 Derivation of the Flexible-Input Share Equation

The establishment's static problem with respect to electricity input is

$$\max_{E_{it}} \; P_{it}\,\mathbb{E}\big[\exp\big(f(k_{it}, l_{it}, m_{it}, e_{it}) + \omega_{it} + \epsilon_{it}\big) \mid k_{it}, l_{it}, m_{it}, e_{it}, \omega_{it}\big] - P_t^E E_{it}, \tag{B16}$$

where $e_{it} \equiv \ln E_{it}$.

The first-order condition is

$$P_{it}\,\mathbb{E}\left[\frac{\partial Y_{it}}{\partial E_{it}} \mid k_{it}, l_{it}, m_{it}, e_{it}, \omega_{it}\right] = P_t^E. \tag{B17}$$

Using

$$\frac{\partial Y_{it}}{\partial E_{it}} = \frac{\partial f(k_{it}, l_{it}, m_{it}, e_{it})}{\partial e_{it}} \frac{Y_{it}}{E_{it}}, \tag{B18}$$

we obtain

$$\frac{P_t^E E_{it}}{P_{it} Y_{it}} = \theta_{E,it}\, \mathbb{E}[\exp(\epsilon_{it}) \mid k_{it}, l_{it}, m_{it}, e_{it}] \exp(-\epsilon_{it}), \tag{B19}$$

where

$$\theta_{E,it} \equiv \frac{\partial f(k_{it}, l_{it}, m_{it}, e_{it})}{\partial e_{it}}. \tag{B20}$$

Imposing the normalization $\mathbb{E}[\exp(\epsilon_{it}) \mid k_{it}, l_{it}, m_{it}, e_{it}] = 1$ yields

$$\ln s_{E,it} = \ln \theta_{E,it} - \epsilon_{it}. \tag{B21}$$

B-3.4 Derivation of CES Distribution Parameters

For the CES aggregator in equation (8), with $\sum_j \alpha_j = 1$ and $\sigma \neq 1$, the associated unit-cost function is proportional to

$$c(p) = \left[\sum_{j \in \{K,L,M,E\}} \alpha_j^\sigma p_j^{1-\sigma} \right]^{1/(1-\sigma)}. \tag{B22}$$

By Shephard's lemma, input j's share of total cost is

$$s_j^C(p) = \frac{\alpha_j^\sigma p_j^{1-\sigma}}{\sum_{k \in \{K,L,M,E\}} \alpha_k^\sigma p_k^{1-\sigma}}. \tag{B23}$$

For the CES technology above, cost minimization implies that an input's normalized output elasticity equals its cost share. We therefore construct the industry-specific cost-share target as

$$\hat{s}_j = \frac{\hat{\theta}_j}{\sum_{k \in \{K,L,M,E\}} \hat{\theta}_k}. \tag{B24}$$

Expressing input prices relative to their reference levels gives $p_j^0 = 1$ for every input. Equation (B23) then implies

$$\hat{s}_j = \frac{\hat{\alpha}_j(\sigma)^\sigma}{\sum_{k \in \{K,L,M,E\}} \hat{\alpha}_k(\sigma)^\sigma}. \tag{B25}$$

Conditional on a candidate σ and imposing $\sum_j \hat{\alpha}_j(\sigma) = 1$, the CES distribution parameters are therefore

$$\hat{\alpha}_j(\sigma) = \frac{\hat{s}_j^{1/\sigma}}{\sum_{k \in \{K,L,M,E\}} \hat{s}_k^{1/\sigma}}, \qquad j \in \{K, L, M, E\}. \tag{B26}$$

Substituting Equation (B26) into the cost-share equation reproduces $\hat{s}_j$ for every candidate value of σ.

C Additional Results and Robustness Checks

C-1 Additional Results

Table C1: Effects of Supply Shocks on Labor

	Dependent variable: $\ln(Y) - \ln(Y_{2010})$				
		Employment			
	Salary per worker (1)	Male (2)	Female (3)	Regular (4)	Non-regular (5)
Panel A: OLS					
Supply shock: 2011–2012	0.00	0.00	0.00	0.02	-0.08
	(0.02)	(0.02)	(0.02)	(0.01)	(0.07)
Supply shock: 2013–2018	-0.03	-0.15***	-0.11**	-0.08	-0.57**
	(0.04)	(0.05)	(0.03)	(0.06)	(0.14)
Panel B: 2SLS					
Supply shock: 2011–2012	0.02	0.00	-0.01	0.03	-0.19*
	(0.02)	(0.02)	(0.02)	(0.02)	(0.07)
Supply shock: 2013–2018	0.00	-0.13**	-0.14	-0.01	-0.93***
	(0.05)	(0.07)	(0.07)	(0.07)	(0.20)
Industry-by-year FE	Yes	Yes	Yes	Yes	Yes
Region clusters	9	9	9	9	9
Obs.: 2011–2012	268,264	268,054	267,338	267,647	229,121
Obs.: 2013–2018	383,863	383,595	382,478	382,953	327,347

Notes: Outcomes are log changes from 2010. See Appendix Section A-1.2 for definitions of worker types. All regressions include three-digit-industry-by-year fixed effects. Phase-out estimates use 2005-2010 and 2011-2012; post-phase-out estimates use 2005-2010 and 2013-2018. Treatment and both instruments are set to zero through 2010. The 2SLS instruments are 2010 nuclear capacity divided by peak demand times common national nuclear-supply growth, and the 2010 reserve margin. Parentheses report standard errors clustered by electricity region. Stars use wild-cluster-bootstrap Wald p-values with Rademacher weights. Observation counts include the pre-period. Stars use unrounded p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

C-2 Electricity Productivity

This section investigates whether electricity supply shocks improved electricity productivity relative to labor productivity. As discussed in the main text, electricity use declined more than labor inputs after the Fukushima accident. This pattern may reflect either higher electricity prices relative to wages or improvements in electricity productivity. Suppose production follows the CES production function in Equation (8). The first-order conditions with respect to electricity and labor imply:

$$\ln\left(\frac{E_{it}}{L_{it}}\right) = -\sigma \ln\left(\frac{p_{it}^E}{p_{it}^L}\right) + (1-\sigma)\,\omega_{it}^E, \tag{C1}$$

where E_{it} and L_{it} denote electricity and labor inputs, p_{it}^E and p_{it}^L are their respective prices, σ is the elasticity of substitution, and ω_{it}^E denotes electricity productivity.

This equation suggests that relative input demand depends on two factors: relative input prices and electricity productivity. Higher electricity prices relative to wages reduce the electricity-to-labor ratio, while changes in electricity productivity affect this ratio even when relative prices are unchanged.

We examine whether higher electricity prices relative to wages explain the decline in electricity use relative to labor input after the Fukushima accident. We consider $\sigma \in \{0.4, 0.6, 0.8\}$. For each value, we recover the change in electricity productivity as

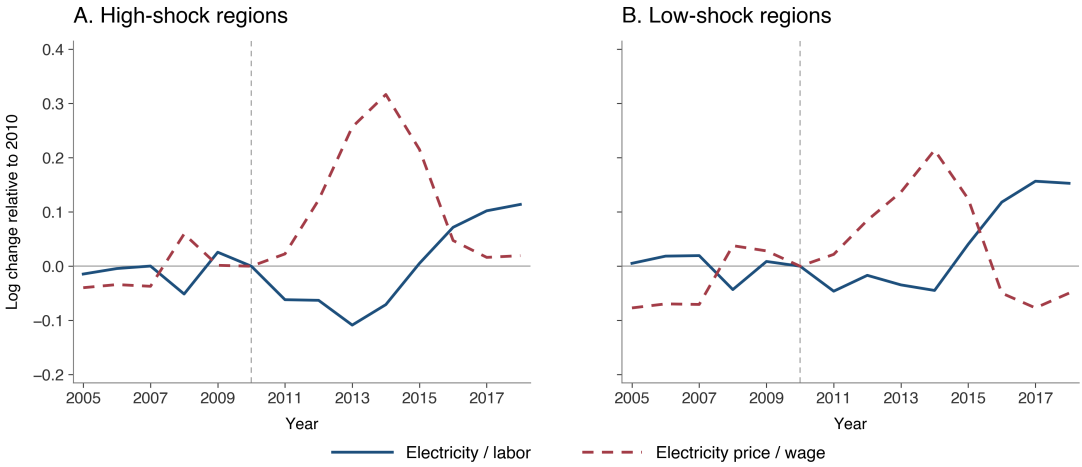
$$\Delta\omega_{it}^E(\sigma) = \frac{\Delta \ln(E_{it}/L_{it}) + \sigma \Delta \ln(p_{it}^E/p_{it}^L)}{1-\sigma}. \tag{C2}$$

This measure captures changes in electricity productivity relative to labor productivity under the CES framework.

Appendix Figure C1 shows how electricity use relative to labor input and electricity prices relative to wages evolved in high- and low-shock regions. Both measures show log changes from their 2010 levels. Relative prices rose more and the electricity-to-labor ratio fell more in high-shock regions (Panel A) than in low-shock regions (Panel B), consistent with substitution away from electricity. Appendix Table C2 reports OLS and 2SLS estimates of the effect of supply shocks on implied electricity productivity.

During 2011–2012, the estimates suggest improvements in relative electricity productivity. For example, the 2SLS coefficient in Column (2), which assumes $\sigma = 0.6$, is 0.42. However, the corresponding estimate falls to 0.06 during 2013–2018. Across the assumed elasticities, the 2013–2018 coefficients vary in sign and range from -1.02 to 0.42, but none is statistically significant. Overall, we find little evidence that supply shocks changed electricity productivity in the medium run.

Figure C1: Relative Electricity Inputs and Prices



Notes: We classify regions with mean supply shocks during 2011–2018 at or above the median across the nine regions as high-shock regions and the remaining regions as low-shock regions. Panel A shows high-shock regions, and Panel B shows low-shock regions. The electricity-to-labor ratio and the price of electricity relative to wages are expressed as log changes relative to 2010.

Table C2: Effects of Supply Shocks on Electricity Productivity

	Dep Var: $\omega_{it}^{E}(\sigma) - \omega_{i,2010}^{E}(\sigma)$		
	$\sigma = 0.4$	$\sigma = 0.6$	$\sigma = 0.8$
	(1)	(2)	(3)
Panel A. OLS			
Supply shock: 2011–2012	0.28***	0.45**	0.96**
	(0.07)	(0.07)	(0.20)
Supply shock: 2013–2018	0.21	-0.11	-1.06
	(0.23)	(0.35)	(0.74)
Panel B. 2SLS			
Supply shock: 2011–2012	0.26	0.42*	0.88*
	(0.13)	(0.09)	(0.20)
Supply shock: 2013–2018	0.42	0.06	-1.02
	(0.38)	(0.63)	(1.46)
Industry-by-year FE	Yes	Yes	Yes
Obs.: 2011–2012	268,264	268,264	268,264
Obs.: 2013–2018	383,863	383,863	383,863

Notes: Regressions include three-digit-industry-by-year fixed effects. Phase-out samples include 2005–2010 and 2011–2012; post-phase-out samples include 2005–2010 and 2013–2018. The 2SLS instruments are 2010 nuclear capacity divided by peak demand times common national nuclear-supply growth, and the 2010 reserve margin. Parentheses report standard errors clustered by electricity region. Stars use wild-cluster-bootstrap Wald p-values with Rademacher weights. Observation counts include the pre-period. Stars use unrounded p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

C-3 Alternative Explanations and Robustness Checks

Table C3: Supply-Chain Exposure Controls

	Dep Var: $\Delta \ln(Y) = \ln(Y) - \ln(Y_{2010})$			
	Baseline (1)	Direct (2)	Total (3)	Both (4)
Panel A. Electricity, OLS				
Supply shock: 2011–2012	-0.14*	-0.12	-0.12	-0.12
	(0.10)	(0.09)	(0.09)	(0.09)
Supply shock: 2013–2018	-0.60**	-0.62**	-0.62**	-0.62**
	(0.15)	(0.16)	(0.16)	(0.16)
Panel B. Electricity, 2SLS				
Supply shock: 2011–2012	-0.15	-0.13	-0.13	-0.13
	(0.18)	(0.14)	(0.14)	(0.14)
Supply shock: 2013–2018	-0.87*	-0.88*	-0.88*	-0.89*
	(0.21)	(0.21)	(0.21)	(0.21)
Panel C. Output, OLS				
Supply shock: 2011–2012	-0.04	-0.02	-0.02	-0.02
	(0.02)	(0.03)	(0.03)	(0.03)
Supply shock: 2013–2018	-0.21***	-0.18**	-0.18**	-0.18**
	(0.04)	(0.04)	(0.04)	(0.04)
Panel D. Output, 2SLS				
Supply shock: 2011–2012	-0.03	-0.02	-0.02	-0.02
	(0.03)	(0.03)	(0.03)	(0.03)
Supply shock: 2013–2018	-0.25**	-0.22**	-0.23**	-0.22**
	(0.07)	(0.05)	(0.05)	(0.05)
Direct exposure × year	No	Yes	No	Yes
Total exposure × year	No	No	Yes	Yes
Industry-by-year FE	Yes	Yes	Yes	Yes
Region clusters	9	9	9	9
Obs.: 2011–2012	268,264	231,695	231,695	231,695
Obs.: 2013–2018	383,863	343,679	343,679	343,679

Notes: Outcomes are log changes from 2010. Direct exposure measures the share of intermediate inputs sourced from prefectures most affected by the earthquake and nuclear accident, while total exposure also captures dependence on these areas through suppliers' suppliers (see Appendix A-3 for definitions). All regressions include three-digit-industry-by-year fixed effects. Phase-out estimates use 2005-2010 and 2011-2012; post-phase-out estimates use 2005-2010 and 2013-2018. Treatment and both instruments are set to zero through 2010. The 2SLS instruments are 2010 nuclear capacity divided by peak demand times common national nuclear-supply growth, and the 2010 reserve margin. Parentheses report standard errors clustered by electricity region. Stars use wild-cluster-bootstrap Wald p-values with Rademacher weights. Observation counts include the pre-period. Stars use unrounded p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C4: Cumulative Local-Demand Controls

	Dep Var: $\Delta \ln(Y) = \ln(Y) - \ln(Y_{2010})$					
	Baseline (1)	GDP (2)	Pop. (3)	Spending (4)	Employ. (5)	Prices (6)
Panel A. Electricity, OLS						
Supply shock: 2011–2012	-0.14	-0.15	-0.14	-0.14	-0.14	-0.14
	(0.11)	(0.11)	(0.11)	(0.11)	(0.10)	(0.10)
Supply shock: 2013–2018	-0.61**	-0.59**	-0.58**	-0.59**	-0.61***	-0.61***
	(0.15)	(0.14)	(0.15)	(0.15)	(0.14)	(0.14)
Panel B. Electricity, 2SLS						
Supply shock: 2011–2012	-0.15	-0.15	-0.15	-0.13	-0.13	-0.13
	(0.18)	(0.17)	(0.17)	(0.18)	(0.17)	(0.17)
Supply shock: 2013–2018	-0.87*	-0.85*	-0.84*	-0.85*	-0.88**	-0.88**
	(0.21)	(0.22)	(0.22)	(0.22)	(0.20)	(0.20)
Panel C. Output, OLS						
Supply shock: 2011–2012	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03
	(0.02)	(0.02)	(0.03)	(0.02)	(0.02)	(0.02)
Supply shock: 2013–2018	-0.21***	-0.21***	-0.20***	-0.21***	-0.21***	-0.21***
	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)
Panel D. Output, 2SLS						
Supply shock: 2011–2012	-0.03	-0.03	-0.02	-0.02	-0.02	-0.02
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Supply shock: 2013–2018	-0.25**	-0.25**	-0.25**	-0.25**	-0.25**	-0.25**
	(0.07)	(0.06)	(0.06)	(0.05)	(0.05)	(0.05)
Economic activity	No	Yes	Yes	Yes	Yes	Yes
Population	No	No	Yes	Yes	Yes	Yes
Local spending	No	No	No	Yes	Yes	Yes
Nonmanufacturing employment	No	No	No	No	Yes	Yes
Consumer prices	No	No	No	No	No	Yes
Industry-by-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region clusters	9	9	9	9	9	9
Obs.: 2011–2012	231,695	231,695	231,695	231,695	231,695	231,695
Obs.: 2013–2018	343,679	343,679	343,679	343,679	343,679	343,679

Notes: Outcomes are log changes from 2010. All regressions include three-digit-industry-by-year fixed effects. Phase-out estimates use 2005-2010 and 2011-2012; post-phase-out estimates use 2005-2010 and 2013-2018. Treatment and both instruments are set to zero through 2010. The 2SLS instruments are 2010 nuclear capacity divided by peak demand times common national nuclear-supply growth, and the 2010 reserve margin. Parentheses report standard errors clustered by electricity region. Stars use wild-cluster-bootstrap Wald p-values with Rademacher weights. Observation counts include the pre-period. Columns (2)–(6) cumulatively add primary- and tertiary-sector GDP, total and working-age population, residential construction and household expenditure, nonmanufacturing employment, and CPI for prefectural capitals excluding food (except alcohol) and energy; all controls enter as log changes from 2010. Stars use unrounded p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C5: Leave-One-Region-Out IV Estimates, 2013–2018

Region omitted	Electricity Use		Output		First-stage F-statistic
	Coefficient	Std. error	Coefficient	Std. error	
	(1)	(2)	(3)	(4)	(5)
None (full sample)	-0.87*	(0.21)	-0.25**	(0.07)	28.2
Hokkaido	-1.01***	(0.15)	-0.27**	(0.07)	26.0
Tohoku	-0.87*	(0.21)	-0.25**	(0.06)	23.4
Tokyo	-0.92	(0.20)	-0.22	(0.06)	27.0
Hokuriku	-0.82	(0.23)	-0.23	(0.09)	28.4
Chubu	-0.70	(0.35)	-0.17	(0.11)	37.5
Kansai	-0.60	(0.24)	-0.32**	(0.08)	6.7
Chugoku	-0.86*	(0.24)	-0.28**	(0.07)	22.5
Shikoku	-0.92*	(0.21)	-0.25*	(0.08)	39.3
Kyushu	-0.91**	(0.18)	-0.26**	(0.07)	24.1

Notes: The table reports IV estimates for 2013-2018. Each exclusion removes one electricity region from both the 2005-2010 and 2013-2018 samples. Outcomes are log changes from 2010. All regressions include three-digit-industry-by-year fixed effects. Phase-out estimates use 2005-2010 and 2011-2012; post-phase-out estimates use 2005-2010 and 2013-2018. Treatment and both instruments are set to zero through 2010. The 2SLS instruments are 2010 nuclear capacity divided by peak demand times common national nuclear-supply growth, and the 2010 reserve margin. Parentheses report standard errors clustered by electricity region. Stars use wild-cluster-bootstrap Wald p-values with Rademacher weights. Observation counts include the pre-period. Stars use unrounded p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D Establishments' Responses to Supply Shocks

This section summarizes our results on the effects of electricity supply shocks on manufacturing establishment behavior.

Product Switching and Outsourcing. Electricity supply shocks had limited effects on product switching and outsourcing, two ways establishments could reduce the electricity intensity of production. Table 4, Panel B, reports 2013–2018 IV coefficients of −0.02 for product switching (Column (1)) and 0.04 for product-level electricity intensity (Column (2)). At the mean shock intensity, these estimates imply only a 0.1 pp decline in product switching and a 0.3% increase in electricity intensity (see Appendix A-1.3 for variable construction). Likewise, the coefficient for outsourcing as a share of materials in Column (3) is small, at −0.01, implying a decline of only 0.1 pp at the mean shock intensity.

Reallocation and Relocation. We find no convincing evidence that electricity supply shocks induced firms to reallocate output across regions or relocate (see Appendix A-1.3 for variable construction). Table D1 examines whether shocks in a region (such as Tokyo) increased electricity use at the same firm's establishments in other regions (non-Tokyo) during 2013–2018. While we find a positive and statistically significant effect on electricity use (0.69; Panel B, Column (1)), the output effect is small and statistically insignificant (0.03; Column (2)), implying only a 0.2% increase at the mean shock of 7.4%. Additionally, electricity supply shocks have little effect on establishment entry and exit (Table D2). Both estimates are statistically insignificant. At the mean shock intensities, these estimates imply that shocks reduce entry by 0.002 ppt and increase exit by 0.10 ppt.[2]

Other Energy Inputs and On-Site Generation. Furthermore, we find no evidence that electricity supply shocks increased establishments' use of other energy inputs in production. Instead, the point estimate suggests a decline: the 2013–2018 IV coefficient for fuel and steam used directly in production is −0.24 (Table 5, Column (1)). Although statistically insignificant, this estimate is similar in magnitude to those for materials (−0.22; Table 3, Panel B, Column (4)) and employment (−0.15; Column (2)), suggesting that other energy inputs contracted alongside other production inputs.

On-site generation helped cushion output losses, consistent with previous studies (Allcott et al., 2016; Cole et al., 2018). In our data, only about 4% of establishments had on-site

[2]We express these changes relative to the 2009 establishment count. We calculate them using the unrounded IV coefficients from Table D2, Panel B, Columns (2) and (3) for entry and exit, respectively, mean shocks of 6.3% for entry and 6.2% for exit, and baseline annualized entry and exit shares of 1.4% and 6.0%, respectively.

generators. Among generator owners, electricity supply shocks increased self-generation by approximately 33% at the mean shock intensity of 7.4%. The point estimates suggest that this increase was large enough, on average, to offset the reduction in electricity purchased from the grid. We calculate this increase using the 2013–2018 IV coefficient of 3.82, which is statistically significant at the 10% level (Table 5, Panel B, Column (3)). Figure D1 also provides suggestive evidence that generators cushioned output losses: we detect no statistically significant output decline among establishments with generators, whereas establishments without generators experienced contractions. However, more than 95% of establishments lacked generators, limiting the scope for this response.

Working Hours. Instead, electricity supply shocks reduced manufacturing working hours, potentially contributing to the decline in output. Table 4, Panel B, Column (4), indicates that a 1 pp increase in shock intensity reduced total working hours per worker by about 0.10% during 2013–2018, implying a 0.8% decline at the mean shock intensity. If a 1% reduction in working hours translates into a 1% reduction in output, this decline could account for about 42% of the estimated 1.8% output loss. Columns (5) and (6) suggest declines in scheduled and overtime hours, with coefficients of -0.09 and -0.15, respectively, although the overtime estimate is imprecise. These findings are consistent with evidence that manufacturers shifted production away from peak periods and, in some cases, curtailed operations to reduce electricity demand after Fukushima (Kimura et al., 2012; Kimura and Nishio, 2016).

Table D1: Spillover Effects: Production Reallocation Across Regions

	Dep Var: $\ln(Y_t) - \ln(Y_{2010})$	
	Electricity	Output
	(1)	(2)
Panel A. OLS		
Supply shock: 2011–2012	0.06	-0.04
	(0.06)	(0.07)
Supply shock: 2013–2018	0.43**	-0.02
	(0.14)	(0.12)
Panel B. 2SLS		
Supply shock: 2011–2012	0.06	-0.07
	(0.07)	(0.09)
Supply shock: 2013–2018	0.69**	0.03
	(0.15)	(0.20)
Industry-by-year FE	Yes	Yes
Regression weights	None	None
Region clusters	9	9
Obs.: 2011–2012	28,780	28,780
Obs.: 2013–2018	40,975	40,975

Notes: Each observation is a firm-region-year. Outcomes are log changes from 2010 in the firm's electricity use and output outside the focal region. The regressor is the focal region's supply shock. Regressions are unweighted and include three-digit-industry-by-year fixed effects. Phase-out estimates use 2005–2010 and 2011–2012; post-phase-out estimates use 2005–2010 and 2013–2018. Parentheses report standard errors clustered by electricity region. Significance levels use wild cluster bootstrap p-values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

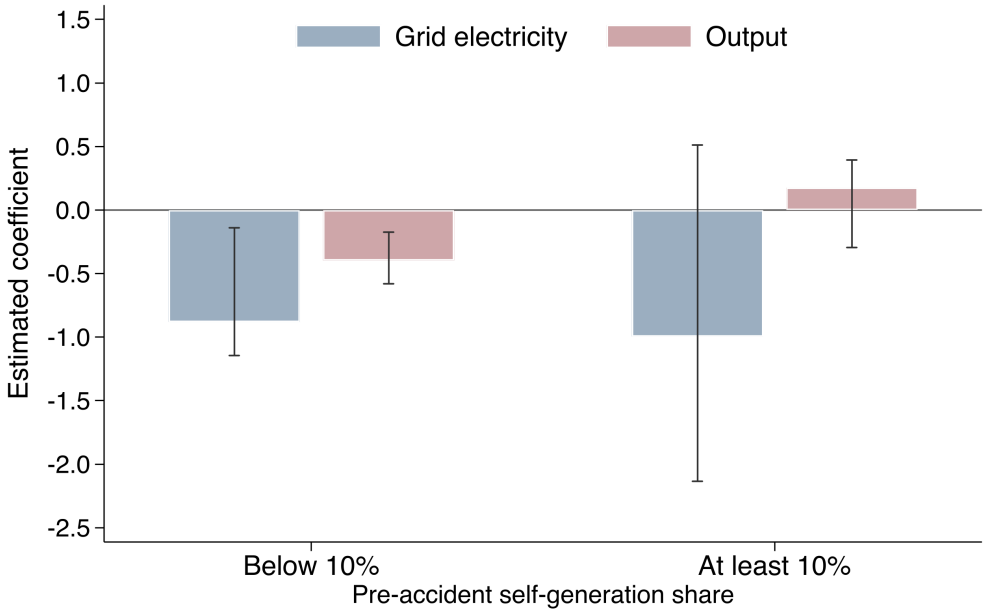
Table D2: Effects of Supply Shocks on Entry and Exit

	Number of establishments	Entry	Exit
	(1)	(2)	(3)
Panel A. OLS			
Supply shock	-0.18	0.52	0.47
	(0.28)	(0.82)	(0.37)
Observations	4,412	3,962	4,258
Panel B. 2SLS			
Supply shock	0.06	-0.02	0.27
	(0.36)	(0.96)	(0.42)
Observations	4,412	3,962	4,258
First-stage F	37.36	52.53	50.00
Industry-by-wave FE	Yes	Yes	Yes
Region clusters	9	9	9

Notes: The sample comprises prefecture-by-industry observations from the 2009, 2012, 2014, and 2016 Economic Censuses. Outcomes are log changes from 2009. Entry and exit counts are annualized and include relocations. Supply shocks and instruments are averaged over the relevant measurement windows. Regressions are unweighted and include two-digit-industry-by-wave fixed effects. First-stage F denotes the Kleibergen–Paap statistic. Parentheses report standard errors clustered by electricity region. Significance levels use wild cluster bootstrap p-values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D-1 Electricity Supply Shocks and On-Site Generation

Figure D1: Effects of Supply Shocks by Generator Ownership



Notes: The figure reports post-phase-out 2SLS estimates for grid electricity purchases and output, measured as log changes from 2010. Regressions use 2005–2010 and 2013–2018 and include three-digit-industry-by-year fixed effects. Vertical bars show 90% confidence intervals from a wild cluster bootstrap clustered by electricity region.

E From Reduced-Form Effects to Production Mechanisms

E-1 Simulation Inputs

Table E1 summarizes the technology, demand, and policy inputs used in the simulation. Three features of the scenario construction are worth noting. First, we hold technology and demand parameters fixed between the no-shock baseline and the post-phase-out scenario. Second, we express all input prices relative to their 2010 levels. Relative non-electricity prices remain one, while the two scenarios differ in the relative electricity price and shock-season rationing.

Table E1: Simulation Inputs

Object	Level	Description or source	Baseline	Post-phase-out
Panel A: Technology				
σ	Common	Elasticity of substitution	$\widehat{\sigma}$	$\widehat{\sigma}$
$\widehat{\boldsymbol{\theta}}_j$	Industry	Estimated output elasticities	$\widehat{\boldsymbol{\theta}}_j$	$\widehat{\boldsymbol{\theta}}_j$
Ω_i	Plant	Baseline productivity	Ω_i	Ω_i
κ	Common	Capital-utilization elasticity	$\widehat{\kappa}$	$\widehat{\kappa}$
v_i	Plant	Baseline capital–electricity ratio; $\frac{K_{2010}}{E_{2010}}$	v_i	v_i
G_i	Plant	Generator ownership	G_i	G_i
Panel B: Demand				
η	Common	Demand elasticity; 5	5	5
B_i	Plant	Demand shifter; $Y_{i,2010}$	B_i	B_i
Panel C: Policies				
R_r	Region	Shock-season electricity-saving target	0	$\overline{R}_r$
ϕ	Common	Shock-season share of the year	-	90/365
p_r^E	Region	Regional electricity price	1	$\overline{p}_r^E$
p^K, p^L, p^M	Common	Non-electricity input prices	1	1

E-2 Predicted Electricity-Price Changes

To construct the shock-induced electricity-price input used in the simulation, we estimate how regional electricity supply shocks affected the prices paid by manufacturing plants. Observed electricity prices increased through 2014 and declined during 2015–2016, largely reflecting nationwide movements in global energy costs. We therefore control for nationwide electricity-price trends to isolate the component attributable to regional supply shocks. We estimate the following specification separately for the phase-out period (2011–2012) and the post-phase-out period (2013–2018):

$$\Delta \ln p_{rt}^E \equiv \ln p_{rt}^E - \ln p_{r,2010}^E = \beta \, \mathcal{S}_{rt} + \gamma_{jt} + \varepsilon_{ijrt}, \tag{E1}$$

where p_{rt}^E denotes the average electricity price in region r and year t, which we assign to plant i; $\mathcal{S}_{rt}$ denotes the region–year supply shock; and γ_{jt} denotes three-digit-industry-by-year fixed

effects. These fixed effects absorb nationwide price movements, including common changes in fossil-fuel costs. We estimate Equation (E1) by OLS and 2SLS, using the instruments defined in Equation (7) for the latter. All specifications use the same 2010 total-salary weights as the main analysis, and standard errors are clustered by region.

Table E2 shows no detectable price response during 2011–2012: both the OLS and 2SLS estimates are close to zero. This pattern is consistent with adjustment lags in regulated tariffs under Japan's cost-of-service regulation. During 2013–2018, the OLS and 2SLS estimates are 0.78 and 1.12, respectively. The 2SLS estimate implies that a 1 ppt increase in the supply shock raises the log electricity price by 0.0112, corresponding to an approximately 1.12% increase.

For the simulation, we normalize the 2010 electricity price to one and use the post-phase-out 2SLS estimate to construct each region's predicted price:

$$\overline{p}_r^E \equiv \frac{p_{r,\text{post}}^E}{p_{r,2010}^E} = \exp\left(\widehat{\beta}_{\text{post}}\,\overline{\mathcal{S}}_r\right), \tag{E2}$$

where $\overline{\mathcal{S}}_r$ is the region's mean supply shock over 2013–2018. The sample's weighted mean shock is 7.4%. The corresponding weighted mean predicted log-price change is 8.4 log points, while the weighted mean of predicted relative prices implies an approximately 9.0% increase in levels. In Tokyo, where the mean shock is 10.8%, the predicted log-price change is 12.9 log points, corresponding to an approximately 13.8% increase in levels. Because regulated tariffs do not vary seasonally within a year, we apply the same predicted price during the shock and off-shock seasons.

Table E2: Effects of Electricity Supply Shocks on Electricity Prices

	Dep Var: $\Delta \ln p^E = \ln p_{rt}^E - \ln p_{r,2010}^E$	
	Phase-out	Post-phase-out
	(1)	(2)
Panel A. The effects of supply shocks using OLS		
Supply shock	0.03	0.78***
	(0.19)	(0.20)
Panel B. The effects of supply shocks using 2SLS		
Supply shock	0.00	1.12**
	(0.32)	(0.35)
Year-by-Industry FE	Yes	Yes

Notes: Column (1) uses data from 2005–2010 and 2011–2012. Column (2) uses data from 2005–2010 and 2013–2018. Panel B instruments the supply shock with predicted nuclear-supply losses and the 2010 reserve margin. All regressions include three-digit-industry-by-year fixed effects. Standard errors clustered by region are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

E-3 Predictions under the Standard CES Technology

We first predict inputs and output under the electricity-supply-shock scenario using the standard CES technology. In this subsection, we abstract from electricity-induced capital idleness by setting $\kappa = 0$. We introduce electricity-dependent capital utilization in Appendix E-4.

E-3.1 Baseline and price-only equilibria

The simulation considers alternative values of the elasticity of substitution, σ. For each candidate value, we recover the CES distribution parameters α using Equation (B14), while holding the baseline cost shares in Equation (B13) fixed.

Plant i in industry j chooses inputs, output, and its output price to maximize profits:

$$\begin{aligned} \max_{K,L,M,E,P,Y} \quad & \Pi_i = P_i Y_i - p^K K_i - p^L L_i - p^M M_i - p^E E_i, \\ \text{subject to} \quad & Y_i = \Omega_i \left(\alpha_{Kj} K_i^{\rho} + \alpha_{Lj} L_i^{\rho} + \alpha_{Mj} M_i^{\rho} + \alpha_{Ej} E_i^{\rho}\right)^{1/\rho}, \\ & Y_i = B_i P_i^{-\eta}, \qquad \rho = \frac{\sigma - 1}{\sigma}. \end{aligned} \tag{E3}$$

For $\sigma \neq 1$, substitution of Equation (B14) into the dual cost function gives

$$c_{ij}(p) = \frac{1}{\Omega_i} \left[\sum_{h \in \{K,L,M,E\}} s_{hj} (p^h)^{1-\sigma} \right]^{1/(1-\sigma)}, \tag{E4}$$

where each p^h is an input-price index relative to its 2010 level. Hence $p^{h,0} = 1$ and $c_{ij}(p^0) = 1/\Omega_i$ for every value of σ. When only the electricity price changes, the relative unit-cost index is

$$\frac{c_{ij}(p^1)}{c_{ij}(p^0)} = \left[1 - s_{Ej} + s_{Ej} (p^{E,1})^{1-\sigma}\right]^{1/(1-\sigma)}. \tag{E5}$$

This expression makes clear that comparisons across σ hold the baseline electricity share fixed. Constant-elasticity demand implies the markup $\mu = \eta/(\eta - 1)$, so the optimal output price and output are

$$P_i^*(p) = \mu c_{ij}(p), \qquad Y_i^*(p) = B_i [\mu c_{ij}(p)]^{-\eta}. \tag{E6}$$

The corresponding conditional factor demands are

$$X_{ih}^*(p) = \frac{Y_i^*(p)}{\Omega_i} \left(\frac{c_{ij}(p) \Omega_i \alpha_{hj}}{p^h} \right)^{\sigma}, \qquad h \in \{K, L, M, E\}. \tag{E7}$$

We express each simulated outcome as the same log difference from the no-shock baseline used in the empirical analysis. Let $p^0 = (1, 1, 1, 1)$ denote the vector of 2010-relative price indices, and let p^1 denote the post-phase-out relative-price vector. Define

$$\Delta \ln c_{ij} \equiv \ln c_{ij}(p^1) - \ln c_{ij}(p^0), \qquad \Delta \ln p^h \equiv \ln p^{h,1} - \ln p^{h,0}. \tag{E8}$$

Because productivity, demand, and the markup are held fixed across simulations, they cancel from the log differences. Taking logs of Equations (E6) and (E7) gives

$$\begin{aligned}
\Delta \ln Y_i &= -\eta \, \Delta \ln c_{ij}, \\
\Delta \ln V_i &= (1-\eta) \, \Delta \ln c_{ij}, \\
\Delta \ln X_{ih} &= (\sigma - \eta) \, \Delta \ln c_{ij} - \sigma \, \Delta \ln p^h, \qquad h \in \{K, L, M, E\}.
\end{aligned} \tag{E9}$$

Here, $V_i \equiv P_i Y_i$ denotes revenue and $\Delta \ln Z_i \equiv \ln Z_i^1 - \ln Z_i^0$ for any outcome Z_i. Thus, we predict the plant-level log changes used in the empirical analysis without using observed plant-level baseline inputs.

For the Cobb–Douglas benchmark, $\sigma = 1$, we evaluate these expressions at the continuous limit of the CES cost function. In this case,

$$\Delta \ln c_{ij} = \sum_{h \in \{K,L,M,E\}} s_{hj} \, \Delta \ln p^h. \tag{E10}$$

E-3.2 Electricity-price elasticities

Define electricity's share in unit cost as

$$s^C_{E,ij}(p) = \frac{s_{Ej} (p^E)^{1-\sigma}}{\sum_{h \in \{K,L,M,E\}} s_{hj} (p^h)^{1-\sigma}}. \tag{E11}$$

At the baseline relative-price vector $p^0 = (1, 1, 1, 1)$, this expression equals s_{Ej} for every σ. By Shephard's Lemma, $\partial \ln c_{ij}(p) / \partial \ln p^E = s^C_{E,ij}(p)$. Combining this result with demand and the conditional factor demands gives

$$\begin{aligned}
\frac{d \ln Y_i}{d \ln p^E} &= -\eta s^C_{E,ij}, \\
\frac{d \ln X_{ih}}{d \ln p^E} &= (\sigma - \eta) s^C_{E,ij}, \qquad h \in \{K, L, M\}, \\
\frac{d \ln E_i}{d \ln p^E} &= -\sigma + (\sigma - \eta) s^C_{E,ij}.
\end{aligned} \tag{E12}$$

Producer surplus is $\Pi_i = (P_i - c_{ij}) Y_i = (\mu - 1) c_{ij} Y_i$, implying

$$\frac{d \ln \Pi_i}{d \ln p^E} = (1 - \eta) s^C_{E,ij}. \tag{E13}$$

E-3.3 Seasonal rationing

Rationing applies only during the shock season S. For a plant in region r, shock-season electricity use must satisfy

$$E_i^S \leq \overline{E}_i^S \equiv (1 - \overline{R}_r) E_i^{S,0}, \tag{E14}$$

where $E_i^{S,0}$ denotes baseline shock-season electricity use. We first solve the unconstrained equilibrium under post-phase-out prices and denote the resulting price-only electricity demand by $E_i^{S,\text{ref}}$.

If $E_i^{S,\text{ref}} \leq \overline{E}_i^S$, rationing is slack and the price-only solution applies. If $E_i^{S,\text{ref}} > \overline{E}_i^S$, we represent the binding constraint with an incremental shadow price. Define

$$\Delta \ln \widetilde{p}_i^E \equiv \ln\left(\frac{\widetilde{p}_i^E}{p_r^{E,1}}\right), \qquad \Delta \ln E_i^{S,R} \equiv \ln\left(\frac{\overline{E}_i^S}{E_i^{S,\text{ref}}}\right). \tag{E15}$$

Using a first-order approximation around the unconstrained post-price equilibrium, the shadow-price adjustment is

$$\Delta \ln \widetilde{p}_i^E = \frac{\Delta \ln E_i^{S,R}}{-\sigma + (\sigma - \eta) s_{E,ij}^C}, \tag{E16}$$

where the electricity cost share is evaluated at the unconstrained post-price equilibrium. The corresponding additional rationing-induced changes are

$$\begin{aligned} \Delta \ln Y_i^{S,R} &= -\eta s_{E,ij}^C \Delta \ln \widetilde{p}_i^E, \\ \Delta \ln X_{ih}^{S,R} &= (\sigma - \eta) s_{E,ij}^C \Delta \ln \widetilde{p}_i^E, \qquad h \in \{K, L, M\}, \\ \Delta \ln \Pi_i^{S,R} &= (1 - \eta) s_{E,ij}^C \Delta \ln \widetilde{p}_i^E. \end{aligned} \tag{E17}$$

The total shock-season response equals the price-only response plus this rationing adjustment. During the off-season O, plants face post-phase-out prices but no rationing.

Annual aggregation. We combine the seasonal predictions using the shock-season share $\phi = 90/365$:

$$\Delta \ln Z_i \equiv \phi \, \Delta \ln Z_i^S + (1 - \phi) \, \Delta \ln Z_i^O. \tag{E18}$$

This expression is a first-order approximation to the log change in an annual outcome constructed by aggregating the two seasons in levels.

E-3.4 Plants with generators

For plants that own generators, we retain the model-simulated response of grid-purchased electricity but set the structural responses of capital, labor, materials, output, and revenue to zero. This is a modeling assumption. The estimates in Section 5.6 are imprecise for generator owners, and the output responses of owners and non-owners are not statistically distinguishable. Those estimates therefore do not establish that owners experienced zero output losses. The ownership adjustment is imposed after the CES solution and is not implied by the first-order conditions above.

E-4 Incorporating Capital Idleness Induced by Electricity Supply Shocks

We use effective capital services, rather than installed capital, to predict inputs and output under electricity supply shocks. Effective capital services are defined in Equation (13) as

$$\widetilde{K}_{it} = \left[(1-\kappa)K_{it}^{\rho} + \kappa(v_i E_{it})^{\rho}\right]^{1/\rho}, \qquad v_i = \frac{K_{i,2010}}{E_{i,2010}}.$$

Because $v_i E_{i,2010} = K_{i,2010}$, effective and installed capital coincide at the 2010 baseline: $\widetilde{K}_{i,2010} = K_{i,2010}$. Moreover, the baseline elasticities of effective capital services with respect to installed capital and electricity are

$$\left.\frac{\partial \ln \widetilde{K}_{it}}{\partial \ln K_{it}}\right|_{t=2010} = 1-\kappa, \qquad \left.\frac{\partial \ln \widetilde{K}_{it}}{\partial \ln E_{it}}\right|_{t=2010} = \kappa. \tag{E19}$$

To incorporate capital idleness, we adjust the estimated capital and electricity output elasticities. Let $\widehat{\theta}_{Kj}$ denote the estimated capital elasticity. Equation (E19) shows that, at the 2010 baseline, a 1% change in installed capital changes effective capital services by $(1-\kappa)$%. Using the chain rule, the adjusted capital elasticity ($\widetilde{\theta}_{Kj}(\kappa)$) is

$$\begin{aligned}
\widetilde{\theta}_{Kj}(\kappa) &\equiv \left.\frac{\partial \ln Y_{it}}{\partial \ln K_{it}}\right|_{t=2010} \\
&= \left.\frac{\partial \ln Y_{it}}{\partial \ln \widetilde{K}_{it}}\frac{\partial \ln \widetilde{K}_{it}}{\partial \ln K_{it}}\right|_{t=2010} \\
&= (1-\kappa)\widehat{\theta}_{Kj}.
\end{aligned} \tag{E20}$$

Electricity affects output directly and through capital idleness. At the 2010 baseline, a 1% change in electricity changes effective capital services by κ%. The adjusted electricity elasticity ($\widetilde{\theta}_{Ej}(\kappa)$) therefore equals the original electricity elasticity plus κ times the capital elasticity:

$$\begin{aligned}
\widetilde{\theta}_{Ej}(\kappa) &\equiv \left.\frac{d \ln Y_{it}}{d \ln E_{it}}\right|_{t=2010} \\
&= \widehat{\theta}_{Ej} + \left.\frac{\partial \ln Y_{it}}{\partial \ln \widetilde{K}_{it}}\frac{\partial \ln \widetilde{K}_{it}}{\partial \ln E_{it}}\right|_{t=2010} \\
&= \widehat{\theta}_{Ej} + \kappa\widehat{\theta}_{Kj}.
\end{aligned} \tag{E21}$$

The output elasticities of labor and materials remain unchanged:

$$\widetilde{\theta}_{Lj}(\kappa) = \widehat{\theta}_{Lj}, \qquad \widetilde{\theta}_{Mj}(\kappa) = \widehat{\theta}_{Mj}. \tag{E22}$$

This adjustment reallocates part of the original capital elasticity to electricity but does not change the sum of the output elasticities.

Using these adjusted output elasticities, we recalculate the industry-specific baseline cost shares as

$$\widetilde{s}_{hj}(\kappa) \equiv \frac{\widetilde{\theta}_{hj}(\kappa)}{\sum_{g\in\{K,L,M,E\}} \widetilde{\theta}_{gj}(\kappa)}, \qquad h \in \{K, L, M, E\}. \tag{E23}$$

Because the sum of the output elasticities is unchanged, these shares can equivalently be written as

$$\begin{aligned} \widetilde{s}_{Kj}(\kappa) &= (1-\kappa)s_{Kj}, & \widetilde{s}_{Lj}(\kappa) &= s_{Lj}, \\ \widetilde{s}_{Mj}(\kappa) &= s_{Mj}, & \widetilde{s}_{Ej}(\kappa) &= s_{Ej} + \kappa s_{Kj}. \end{aligned} \tag{E24}$$

Finally, for each candidate value of σ, we recover the CES distribution parameters from these adjusted baseline shares following Equation (B14). Thus, κ changes the baseline allocation of the capital and electricity shares.

E-4.1 Minimum-Distance Estimation

We jointly estimate the elasticity of substitution, σ, and the capital-utilization elasticity, κ, by minimizing the distance between the model-implied changes and their reduced-form counterparts. We first construct the normalized baseline shares in Equation (B13). For each candidate κ, we form the effective shares in Equation (E23); for each candidate σ, we map those fixed shares into CES coefficients using Equation (B14). We then obtain annual plant-level changes in electricity, labor, materials, and output using the simulation procedure described in Appendix E-3. We aggregate these predictions using the same baseline salary weights as in the main reduced-form regressions. For each outcome, the empirical target is the post-phase-out 2SLS coefficient multiplied by the weighted mean supply shock. The moment discrepancies are

$$g_q(\sigma,\kappa) \equiv \frac{\sum_i w_i \Delta \ln q_i^{\text{sim}}(\sigma,\kappa)}{\sum_i w_i} - \widehat{\beta}_q^{\text{RF}} \overline{\mathcal{S}}_w, \qquad q \in \{E, L, M, Y\}, \tag{E25}$$

where w_i denotes the corresponding regression weight and $\overline{\mathcal{S}}_w \equiv \sum_i w_i \mathcal{S}_i / \sum_i w_i$ is the weighted mean supply shock.

Using an identity weighting matrix, we define the minimum-distance criterion as

$$Q(\sigma,\kappa) = \sum_{q\in\{E,L,M,Y\}} g_q(\sigma,\kappa)^2. \tag{E26}$$

The estimator is

$$(\widehat{\sigma},\widehat{\kappa}) = \underset{\sigma>0,\, 0\le\kappa\le 1}{\arg\min}\; Q(\sigma,\kappa). \tag{E27}$$

We first evaluate $Q(\sigma,\kappa)$ over a coarsely spaced grid of candidate parameter values to identify

the region containing the minimum and then evaluate it over a more finely spaced grid within that region to obtain $(\widehat{\sigma}, \widehat{\kappa})$.

E-4.2 Nested CES Robustness

The baseline model uses a common substitution elasticity for the effective-capital composite and the remaining inputs. We examine whether the results change when electricity and capital services have a separate, lower within-nest elasticity. Let σ_{EK} denote the elasticity between installed capital and the electricity services used to operate it, and let σ_O denote the outer elasticity between the resulting composite, labor, materials, and direct electricity services. Holding the baseline inner shares fixed, the price index for the electricity–capital composite is

$$c_{EK}(p^E; \sigma_{EK}, \kappa) = \left[(1-\kappa) + \kappa (p^E)^{1-\sigma_{EK}}\right]^{1/(1-\sigma_{EK})}. \tag{E28}$$

For $\sigma_{EK} = 0$, we evaluate Equation (E28) at its Leontief limit,

$$c_{EK}(p^E; 0, \kappa) = (1-\kappa) + \kappa p^E.$$

This dual formulation avoids evaluating the numerically unstable primal CES exponent when σ_{EK} is close to zero.

Using the normalized production-function elasticities as baseline outer shares, the outer unit-cost index is

$$c_j(p^E) = \left[s_{Kj} c_{EK}(p^E)^{1-\sigma_O} + s_{Lj} + s_{Mj} + s_{Ej} (p^E)^{1-\sigma_O}\right]^{1/(1-\sigma_O)}. \tag{E29}$$

Thus, changing either elasticity changes the curvature of the technology but not its baseline shares. For each imposed value $\sigma_{EK} \in \{0, 0.10, 0.30, 0.50, 0.70\}$, we recover the corresponding inner and outer CES coefficients and re-estimate (σ_O, κ) using the minimum-distance moments in Equation (E25), subject to $\sigma_O \leq 1$. We focus on this constrained calibration because the nested specification treats the electricity–capital composite as weakly complementary to labor and materials.

Table E3: Nested CES Calibration by Within-Nest Elasticity

	Outer complementarity: $\sigma_O \leq 1$		Predicted changes				RMSE
σ_{EK}	$\widehat{\sigma}_O$	$\widehat{\kappa}$	$\Delta \ln E$	$\Delta \ln L$	$\Delta \ln M$	$\Delta \ln Y$	
0.00	1.00†	0.20	-6.10	-1.47	-1.47	-1.84	0.344
0.10	1.00†	0.19	-6.42	-1.39	-1.39	-1.73	0.168
0.30	0.88	0.18	-6.74	-1.35	-1.35	-1.63	0.058
0.50	0.66	0.17	-6.75	-1.36	-1.36	-1.57	0.090
0.70	0.45	0.16	-6.75	-1.38	-1.38	-1.51	0.120
Reduced-form target			-6.72	-1.29	-1.29	-1.72	

Notes: The table fixes the within-nest electricity–capital elasticity, σ_{EK}, at the value in the first column and re-estimates the outer elasticity and capital-utilization parameter using the same identity-weighted minimum-distance criterion as the baseline model, subject to $\sigma_O \leq 1$. Predicted changes and RMSE are multiplied by 100 and are therefore reported in log points. A dagger indicates that the outer-elasticity estimate is at its upper boundary. The reduced-form targets equal the post-phase-out 2SLS coefficients evaluated at the 2010-total-salary-weighted mean supply shock. Generator owners retain their simulated grid-electricity response, while their non-electric inputs and output remain at baseline, as in the main simulation.

Under outer complementarity, the estimated outer elasticity reaches its upper boundary for the Leontief and 0.10 within-nest specifications, and these cases fit the reduced-form targets less closely. The estimates are interior for $\sigma_{EK} \in \{0.30, 0.50, 0.70\}$, with $\widehat{\sigma}_O$ declining from 0.88 to 0.45 and $\widehat{\kappa}$ remaining between 0.16 and 0.18. Among the displayed values, $\sigma_{EK} = 0.30$ provides the closest fit while preserving complementarity in both nests. Without the outer-complementarity restriction, the Leontief and 0.10 specifications instead require $\widehat{\sigma}_O = 1.23$ and 1.11, respectively. We therefore treat $\sigma_{EK} = 0.30$ as a representative nested-CES specification and interpret the remaining values as sensitivity checks.

E-5 Results for Elasticities

Table E4: Comparison of Elasticity Estimates

	Estimation method	
	Reduced-form (1)	Structural model (2)
Own-price elasticity (electricity)	-0.80	-0.74
Cross-price elasticity (labor)	-0.15	-0.15
Cross-price elasticity (materials)	-0.15	-0.15
Output elasticity	-0.20	-0.17
Preferred parameters	$\sigma = 0.57$, $\kappa = 0.17$	

Notes: The table reports elasticities of inputs and output with respect to electricity-price changes. Column (1) divides each main reduced-form IV coefficient by the estimated IV response of log electricity prices to the supply shock. This reduced-form ratio does not explicitly incorporate the model-implied shadow price of a binding rationing constraint: the outcome coefficient may reflect both price and rationing responses, whereas the denominator contains only the market-price response. Column (2) reports finite-change structural elasticities from the price-only simulation under $\widehat{\sigma} = 0.57$ and $\widehat{\kappa} = 0.17$. For each plant, we divide its predicted log outcome response by its predicted log market-price change and then take the 2010-total-salary-weighted mean; rationing and its shadow price are inactive by construction. Generator owners retain their simulated grid-electricity response, while their structural labor, materials, and output responses are set to zero.

F Counterfactual Analysis

F-0.1 Market integration and reserve sharing

Integrated-market pricing. We construct counterfactual electricity prices under market integration by specifying short-run regional supply curves, aggregating supply across regions, and solving for the integrated-market price. We parameterize the inverse supply curve in constant-elasticity form:

$$p_{rt} = \left(\frac{Q_{rt}}{c_{rt}}\right)^{\beta}, \tag{F1}$$

where p_{rt} is the electricity price, Q_{rt} is quantity supplied, $c_{rt} > 0$ is a region–year cost shifter, and $\beta \equiv d\ln p / d\ln Q$ (so $d\ln Q / d\ln p = 1/\beta$).[3]

Under full integration, all regions face a common price p_t^* in year t, implying regional supply

$$Q_{rt}^* = c_{rt}\,(p_t^*)^{1/\beta}. \tag{F2}$$

Holding fixed national demand $\bar{Q}_t$, the integrated equilibrium satisfies

$$\bar{Q}_t = \sum_r Q_{rt}^* = (p_t^*)^{1/\beta} \sum_r c_{rt}, \tag{F3}$$

which yields

$$p_t^* = \left(\frac{\bar{Q}_t}{\sum_r c_{rt}}\right)^{\beta}. \tag{F4}$$

Reserve sharing (effective capacity). Integration can reduce the need for region-specific reserve margins. We capture this mechanism by scaling aggregate effective capacity by $(1+\kappa)$:

$$p_t^* = \left(\frac{\bar{Q}_t}{(1+\kappa)\sum_r c_{rt}}\right)^{\beta}. \tag{F5}$$

We set $\kappa = 0.007$, corresponding to a 0.7% increase in effective capacity due to improved dispatch and reserve sharing.

Estimation. We set $\beta = 5$, corresponding to a supply elasticity $d\ln Q / d\ln p = 0.2$.[4] Rewrite (F1) in logs:

$$\ln p_{rt} = A_{rt} + \beta \ln Q_{rt}, \qquad A_{rt} \equiv -\frac{1}{\beta}\ln c_{rt}.$$

[3] See Wolak (2000), Bushnell and Mansur (2005), Holland and Mansur (2006), and Cicala (2022).

[4] Existing estimates of very short-run electricity supply elasticities are typically smaller (0.02–0.10) (Wolak, 2000; Bushnell and Mansur, 2005; Holland and Mansur, 2006; Cicala, 2022). Because our variation is at the region–year level, we adopt an intermediate value that allows for limited within-year adjustment.

We recover $A_{rt} = \ln p_{rt} - \beta \ln Q_{rt}$ and model it as a function of generation shares and fixed effects:

$$A_{rt} = \gamma_1 Sh_{rt}^{\text{Nuclear}} + \gamma_2 Sh_{rt}^{\text{Hydro}} + \gamma_3 Sh_{rt}^{\text{Thermal}} + \gamma_4 Sh_{rt}^{\text{Others}} + \tau_t + \delta_r + \varepsilon_{rt}. \tag{F6}$$

We map the fitted value $\widehat{A}_{rt}$ back into the cost shifter via

$$\widehat{c}_{rt} = \exp\Big(-\widehat{A}_{rt}/\beta\Big), \tag{F7}$$

and substitute $\widehat{c}_{rt}$ into (F5) to obtain the integrated-market counterfactual price p_t^*.

Results. Table F1, Column (2), reports implied price changes under market integration. High-cost regions experience sizable counterfactual price reductions, while low-cost regions experience price increases, reflecting reallocation of generation toward lower-cost supply. The 2010-total-salary-weighted national average log-price change falls from 8.4 log points in the baseline to 5.7 log points under integration, corresponding to level increases of approximately 9.0% and 5.8%, respectively.

Table F1: Regional Price Changes: Baseline and Market Integration

		$100 \times \Delta \ln p^E$	
#	Region	Baseline (1)	Market integration (2)
1	Hokkaido	1.1%	-12.6%
2	Tohoku	8.7%	1.7%
3	Tokyo	12.2%	3.1%
4	Hokuriku	4.3%	18.7%
5	Chubu	3.6%	5.0%
6	Kansai	12.9%	8.0%
7	Chugoku	3.6%	12.8%
8	Shikoku	5.0%	2.3%
9	Kyushu	6.4%	10.1%
	National	8.4%	5.7%

Notes: Entries report $100 \times \Delta \ln p^E$. Column (1) reports the shock-induced price increase (baseline), and Column (2) reports the counterfactual price change under market integration. National values are weighted averages over the current plant sample using 2010 total salary.

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